

Cisco 642-874



**Designing Cisco Network Service Architectures
(ARCH) v2.1
Version: 6.6**

Topic 1, Volume A

QUESTION NO: 1

Which of these Layer 2 access designs does not support VLAN extensions?

- A. FlexLinks
- B. loop-free U
- C. looped square
- D. looped triangle

Answer: B

Explanation:

QUESTION NO: 2

As a critical part of the design for the Enterprise Campus network, which of the following two are true concerning intrusion detection and prevention solution? (Choose two)

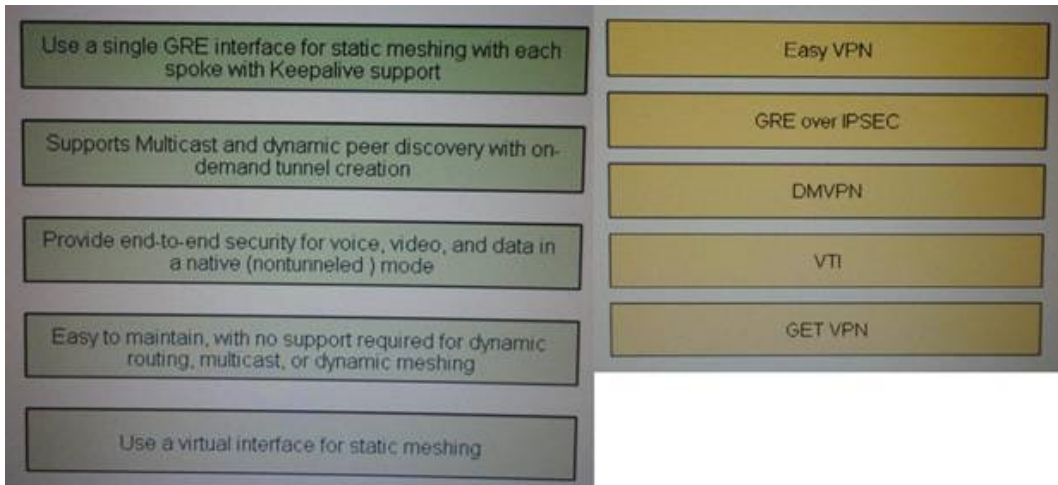
- A. IDS is capable of both inline and promiscuous monitoring, while IPS is only capable of promiscuous monitoring
- B. IDS will stop malicious traffic from reaching its intended target for certain types of attacks.
- C. IPS processes information on Layers 3 and 4 as well as analyzing the contents and payload of the packets for more sophisticated embedded attacks (Layers 3 to 7)
- D. IPS inspects traffic statefully and needs to see both sides of the connection to function properly
- E. IDS placement at the perimeter of Data Center outside the firewall generates many warnings that have relatively low value because no action is likely to be taken on this information

Answer: C,D

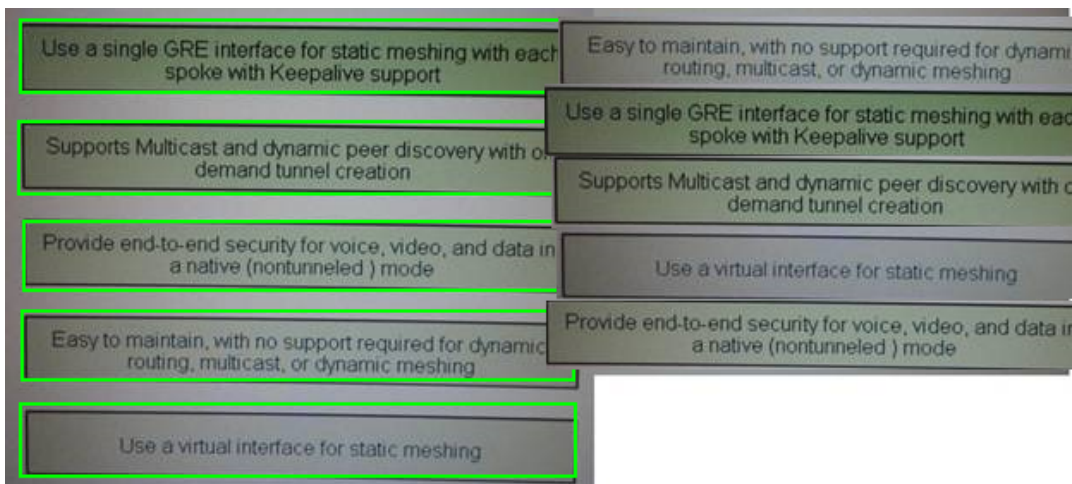
Explanation:

QUESTION NO: 3 DRAG DROP

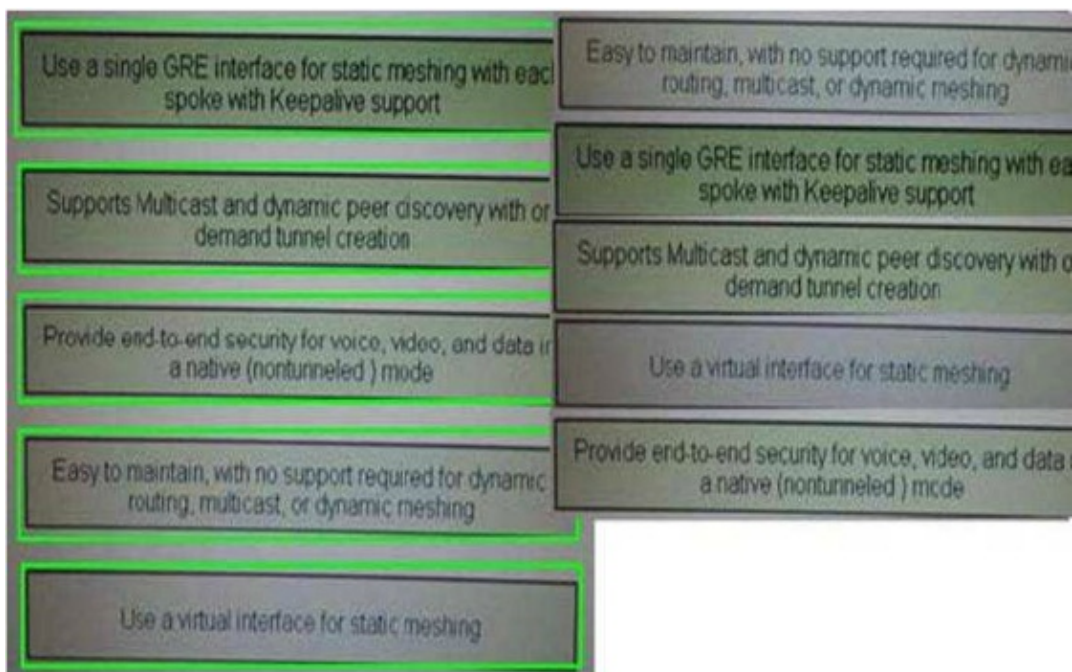
Drag the characteristic on the left to the corresponding IPSec VPN solution on the right.



Answer:



Explanation:



QUESTION NO: 4

OSPF stub areas are an important tool for the Network designer; which of the following two should be considered when utilizing OSPF stub areas? (Choose two)

- A. OSPF stub areas increase the size of the LSDB with the addition of Type 3 and 5 LSAs
- B. OSPF not so stubby areas are particularly useful as a simpler form of summarization
- C. OSPF stub areas are always insulated from external changes
- D. OSPF stub areas can distinguish among ASBRs for destinations that are external to the OSPF domain
- E. OSPF totally stubby areas cannot distinguish among ABRs for the best route to destinations outside the area

Answer: C,E

Explanation:

QUESTION NO: 5

Which two statements are correct regarding Flex Links? (Choose two)

- A. An interface can belong to multiple Flex Links.
- B. Flex Links operate only over single pairs of links.
- C. Flex Link pairs must be of the same interface type
- D. Flex Links automatically disable STP so no BPDUs are propagated
- E. Failover from active to standby on Flex Links takes less than a second

Answer: B,D

Explanation:

QUESTION NO: 6

Which of these technologies is characterized as being a multipoint Layer 2 VPN that connects two or more customer devices using Ethernet bridging techniques?

- A. DPT
- B. MPLS
- C. VPLS
- D. CWDM
- E. DWDM
- F. SONET/SDH

Answer: C

Explanation:

QUESTION NO: 7 DRAG DROP

Drag the best practice recommendation for an Enterprise Campus network on the left to the technology to which it most applies on the right.

use specifically on fiber-optic interconnections that link switches	STP
ensure that an individual link failure will not result in an STP failure	Trunks
enable specifically at the network edge	UDLD
always use a number of links that is a power of 2 to optimize the load balancing of traffic	Etherchannel
manually prune unused VLANs	VSS

Answer:

use specifically on fiber-optic interconnections that link switches	enable specifically at the network edge
ensure that an individual link failure will not result in an STP failure	manually prune unused VLANs
enable specifically at the network edge	use specifically on fiber-optic interconnections that link switches
always use a number of links that is a power of 2 to optimize the load balancing of traffic	ensure that an individual link failure will not result in an STP failure
manually prune unused VLANs	always use a number of links that is a power of 2 to optimize the load balancing of traffic

Explanation:

STP – Enabled specifically on network edge
 Trunks – Manually prune VLANs
 UDLD – Used specifically on Fiber-Optic Interconnection
 Etherchannel – Ensure that an individual link failure will not result in STP Failure
 VSS – Always use a no of links that is power of 2

QUESTION NO: 8

Why is STP required when VLANs span access layer switches?

- A. to ensure a loop-free topology
- B. to protect against user-side loops
- C. in order to support business applications
- D. because of the risk of lost connectivity without STP
- E. for the most deterministic and highly available network topology

Answer: B

Explanation:

QUESTION NO: 9

When designing the IP routing for the Enterprise Campus network, which of the following two iBGP considerations should be taken into account? (Choose two)

- A. iBGP dual homing with different ISPs puts the Enterprise at risk of becoming a transit network

- B. iBGP requires a full mesh of eBGP peers
- C. Routers will not advertise iBGP learned routes to other iBGP peers
- D. The use of route reflectors or Confederations eliminate any full mesh requirement while helping to scale iBGP
- E. iBGP peers do not add any information to the AS path.

Answer: C,E

Explanation:

QUESTION NO: 10

Which virtualization technology allows multiple physical devices to be combined into a single logical device?

- A. device visualization
- B. device clustering
- C. server visualization
- D. network visualization

Answer: B

Explanation:

QUESTION NO: 11

Which two of these are characteristics of MPLS VPNs? (Choose two)

- A. Layer 3 MPLS VPNs can forward only IP packets
- B. Layer 2 MPLS VPNs can forward any network protocol
- C. MPL S label paths are automatically formed based on Layer 2 frames
- D. Layer 3 MPLS VPNs can forward any network protocol based on Layer 2 frames
- E. In Layer 2 MPLS VPNS, the service provider controls the customer Layer 3 policies

Answer: A,B

Explanation:

QUESTION NO: 12

Which technology is an example of the need for a designer to clearly define features and desired performance when designing advanced WAN services with a service provider?

- A. FHRP to remote branches.
- B. Layer 3 MPLS VPNs secure routing
- C. Control protocols (for example Spanning Tree Protocol) for a Layer 3 MPLS service.
- D. Intrusion prevention, QoS, and stateful firewall support network wide.

Answer: B

Explanation:

QUESTION NO: 13

Which three routing protocols can minimize the number of routes advertised in the network?
(Choose three)

- A. IGRP
- B. RIPv2
- C. OSPF
- D. EIGRP
- E. BGP

Answer: B,C,D

Explanation:

QUESTION NO: 14

There are 3 steps to confirm whether a range of IP address can be summarized. When of the following is used in each of these 3 steps?

- A. The first number in the contiguous block of addresses
- B. The last number in the contiguous block of addresses
- C. The size of the contiguous block of addresses
- D. The subnet mask of the original network address

Answer: C

Explanation:

QUESTION NO: 15 DRAG DROP

Drag the characteristic on the left the associated firewall deployment or topology on the right.

Virtual firewall contexts are configured in active/standby pairs on different physical units	Transparent firewall mode
Connected to different service providers and the outbound connection does not use a NAT address	Zone-based policy firewalls
A virtual firewall with its own policies such as NAT, ACLs, and protocol fix-ups	Firewall Contexts
Can use an EtherType ACLs to allow non-IP traffic	Active-active firewall topology
Traffic is subjected to policy restrictions as it crosses regions across the security borders of the network	Asymmetric routing

Answer:

Virtual firewall contexts are configured in active/standby pairs on different physical units	Can use an EtherType ACLs to allow non-IP traffic
Connected to different service providers and the outbound connection does not use a NAT address	Traffic is subjected to policy restrictions as it crosses regions across the security borders of the network
A virtual firewall with its own policies such as NAT, ACLs, and protocol fix-ups	A virtual firewall with its own policies such as NAT, ACLs, and protocol fix-ups
Can use an EtherType ACLs to allow non-IP traffic	Virtual firewall contexts are configured in active/standby pairs on different physical units
Traffic is subjected to policy restrictions as it crosses regions across the security borders of the network	Connected to different service providers and the outbound connection does not use a NAT address

Explanation:

Virtual firewall contexts are configured in active/standby pairs on different physical units Active-active firewall topology
 Connected to different service providers and the outbound connection does not use a NAT address Asymmetric routing
 A virtual firewall with its own policies such as NAT, ACLs and protocol fix-ups Firewall Contexts
 Can use an EtherType ACLs to allow non-IP traffic Transparent firewall mode
 Traffic is subjected to policy restrictions as it crosses regions across the security borders of the network Zone-based policy firewalls

QUESTION NO: 16

Which of these recommendations is most appropriate for the core layer in the Cisco Campus Architecture?

- A. Utilize Layer 3 switching
- B. Utilize software accelerated services
- C. Aggregate end users and support a feature-rich environment
- D. Perform packet manipulation and filtering at the core layer
- E. Use redundant point to-point Layer 2 interconnections when where is a link or node failure.

Answer: A

Explanation:

QUESTION NO: 17

Which of the following is true concerning best design practices at the switched Access layer of the traditional layer2 Enterprise Campus Network?

- A. Cisco NSF with SSO and redundant supervisors has the most impact on the campus in the Access layer
- B. Provides host-level redundancy by connecting each end device to 2 separate Access switches
- C. Offer default gateway redundancy by using dual connections from Access switches to redundant Distribution layer switches using a FHRP
- D. Include a link between two Access switches to support summarization of routing information

Answer: A

Explanation:

QUESTION NO: 18

Which three Layer 2 access designs have all of their uplinks in an active state? (Choose three.)

- A. Flex Links

- B. loop-free U
- C. looped square
- D. looped triangle
- E. loop-free inverted U

Answer: B,C,E

Explanation:

QUESTION NO: 19

In base e-Commerce module designs, where should firewall perimeters be placed?

- A. core layer
- B. Internet boundary
- C. aggregation layer
- D. aggregation and core layers
- E. access and aggregation layers

Answer: A

Explanation:

QUESTION NO: 20

When an Enterprise Campus network designer is addressing the merger of two companies with different IGP, which of the following is considered a superior routing design?

- A. Eliminate the management and support for redistribution by choosing and cutting over to a single IGP at the time of merger
- B. Maintain distinct pockets across a moving boundary of routing protocols, redistributing between them
- C. Manipulate the administrative distance of the different IGPs to be equal throughout the network
- D. Leave the IGPs independent without redistribution wherever communication between company entities is not required

Answer: B

Explanation:

QUESTION NO: 21

From a design perspective which two of the following OSPF Statements are most relevant?
(Choose two)

- A. OSPF stub areas can be thought of as a simple form of summarization
- B. OSPF cannot filter intra area routes
- C. An ARR can only exist in two areas - the backbone and one adjacent area
- D. Performance issues in the Backbone area can be offset by allowing some traffic to transit a non-backbone area
- E. the size of an area (the LSDB) will be constrained by the size of the IP MTU

Answer: A,B

Explanation:

QUESTION NO: 22

When is a first-hop redundancy protocol needed in the distribution layer?

- A. when the design implements Layer 2 between the access and distribution blocks
- B. when multiple vendor devices need to be supported
- C. when preempt tuning of the default gateway is needed
- D. when a robust method of backing up the default gateway is needed
- E. when the design implements Layer 2 between the access switch and the distribution blocks
- F. when the design implements Layer 2 between the access and distribution blocks

Answer: F

Explanation:

QUESTION NO: 23

Which two statements about layer 3 access designs are correct? (Choose two.)

- A. IP address space is difficult to manage.
- B. Broadcast and fault domains are increased
- C. Convergence time is fractionally slower than STP
- D. Limits on clustering and NIC teaming are removed
- E. Fast uplink convergence is supported for failover and fallback

Answer: A,E

Explanation:

QUESTION NO: 24

Which two statements about SCSI are true? (Choose two)

- A. The bus is limited to 32 devices
- B. It is a full duplex serial standard
- C. It is a half-duplex serial standard
- D. It allows up to 320 MB/s of shared channel bandwidth

Answer: C,D

Explanation:

QUESTION NO: 25

What are two characteristics of Server Load Balancing router mode? (Choose two)

- A. The design supports multiple server subnets
- B. An end-user sees the IP address of the real server
- C. SLB routes between the outside and inside subnets
- D. The source or destination MAC address is rewritten, but the IP addresses left alone
- E. SLB acts as a "bump in the wire" between servers and upstream firewall or Layer 3 devices

Answer: A,C

Explanation:

QUESTION NO: 26

When designing the Network Admission Control (NAC) Appliance for the Enterprise Campus Network, which of the following requirements would help the designer to narrow down the NAC choices, from Virtual Gateway to Real IP Gateway, or from In-band to out-of-band?

- A. QoS ToS/DSCP values are required to be forwarded transparently
- B. Device redundancy is required

- C. Per-user ACL support is required
- D. Multicast service support is required

Answer: D

Explanation:

QUESTION NO: 27 DRAG DROP

Drag the characteristic on the left to the corresponding RP model on the right.

Dynamic, utilizing link-local multicast messages which are flooded hop by hop	Static RP
Static, with fault tolerance utilizing with the Multicast Source Discovery Protocol	Auto-RP
Static, with no inherent failover or load balancing mechanism	Anycast RP
Dynamic, utilizing RP mapping agents with dense mode flooding	BSR

Answer:

Dynamic, utilizing link-local multicast messages which are flooded hop by hop	Static, with no inherent failover or load balancing mechanism
Static, with fault tolerance utilizing with the Multicast Source Discovery Protocol	Dynamic, utilizing RP mapping agents with dense mode flooding
Static, with no inherent failover or load balancing mechanism	Static, with fault tolerance utilizing with the Multicast Source Discovery Protocol
Dynamic, utilizing RP mapping agents with dense mode flooding	Dynamic, utilizing link-local multicast messages which are flooded hop by hop

Explanation:

Static RP – Static with no inherent
 Auto RP – Dynamic utilizing RP mapping
 Anycast – Static with
 fault tolerance
 BSR – Dynamic Utilizing Link-local

QUESTION NO: 28

During consultation, you find that a customer has only a single asset closet and is looking for a solution that is easy to deploy. Which NAS physical deployment model would you suggest to this customer?

- A. edge
- B. central
- C. Layer 2
- D. Layer 3

Answer: A

Explanation:

QUESTION NO: 29

How does the Ethernet Relay Service use the VLAN tag?

- A. to provide service internetworking
- B. to support transparency for Layer 2 frames
- C. as a connection identifier to indicate destination
- D. as a mapping to the DLCI in service internetworking
- E. to provide a trunk by which all VLANs can navigate from one site to one or multiple sites

Answer: C

Explanation:

QUESTION NO: 30

What is the most common mode for a firewall?

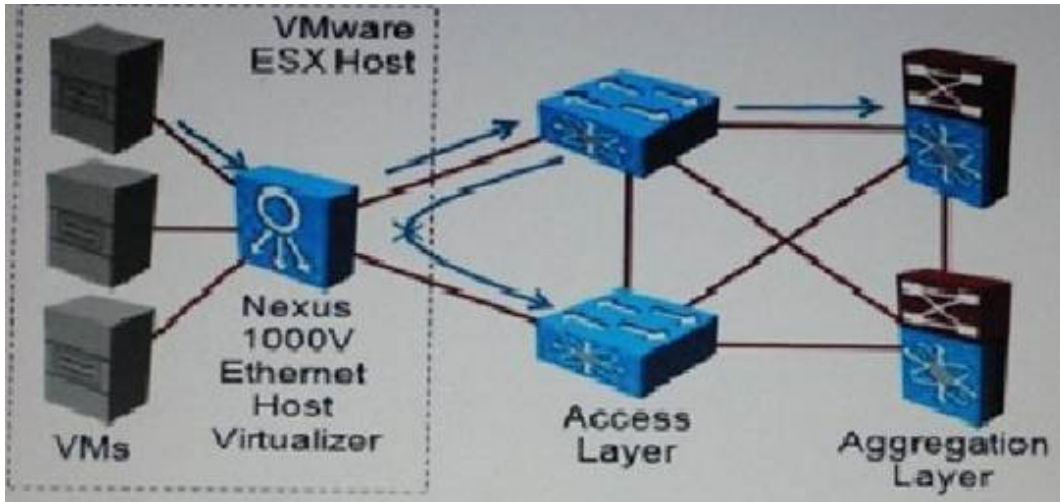
- A. routed mode
- B. context mode
- C. bridged mode
- D. transparent mode
- E. full security mode

Answer: A

Explanation:

QUESTION NO: 31

Refer to the exhibit.



The Cisco Nexus 1000V in the VMware vSphere solution effectively creates an additional access layer in the virtualized data center network; which of the following 1000V characteristics can the designer take advantage of?

- A.** Offloads the STP requirement from the external Access layer switches
- B.** If upstream access switches do not support vPC or VSS the dual-homed ESX host traffic can still be distributed using virtual port channel host mode using subgroups automatically discovered through CDP
- C.** Allow transit traffic to be forwarded through the ESX host between VMNICs
- D.** Can be divided into multiple virtual device contexts for service integration, enhanced security, administrative boundaries, and flexibility of deployment

Answer: B

Explanation:

QUESTION NO: 32

What two descriptions best define DWDM? (Choose two)

- A.** a WDM system that is compatible with EDFA technology
- B.** an optical technology for transmitting up to 16 channels over multiple fiber strands
- C.** an optical technology for transmitting up to 32 channels over multiple fiber strands

- D.** a technology for transmitting multiple optical signals using less sophisticated transceiver design than CWDM
- E.** a technology for transmitting more closely packed optical signals using more sophisticated transceiver designs than CWDM

Answer: A,E

Explanation:

QUESTION NO: 33

Which two characteristics are true of IVRs? (Choose two)

- A.** They are known as fabric routing
- B.** They cannot span multiple switches
- C.** Their connectivity is supported by Layer 2
- D.** They enable devices in different VSAN fabrics to communicate
- E.** They require that multiple switch fabrics be merged before they can function

Answer: A,D

Explanation:

QUESTION NO: 34

Which of these is a correct description of Stateful Switchover?

- A.** It will only become active after a software failure
- B.** It will only become active after a hardware failure
- C.** It requires that Cisco N3F be enabled in order to work successfully
- D.** It synchronizes the MAC, FIB, and adjacency tables between Active and Standby Route Processors.

Answer: D

Explanation:

QUESTION NO: 35

Which technology is best suited for the most scalable means to separate the data plane for a

Layer3 VPN?

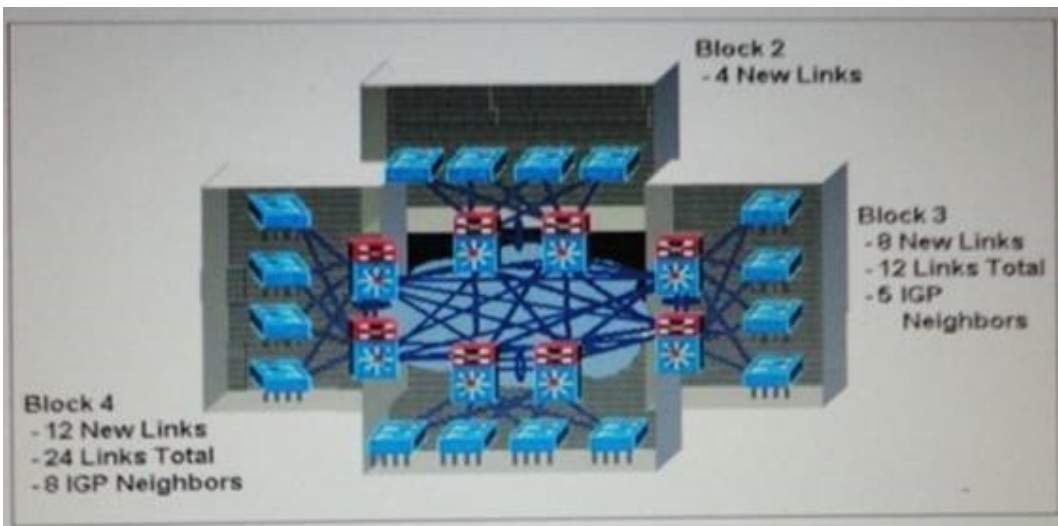
- A. GRE
- B. 802.1Q
- C. MPLS
- D. L2TPv3

Answer: C

Explanation:

QUESTION NO: 36

Refer to the exhibit.



Which recommended practice is applicable?

- A. If no core layer is deployed, the design will be easier to scale
- B. A dedicated campus core layer should be deployed for connecting three or more buildings
- C. If no core layer is deployed, the distribution switches should not be fully meshed
- D. A dedicated campus core layer is not needed for connecting fewer than five buildings

Answer: B

Explanation:

A dedicated campus core layer should be deployed for connecting 3 or more buildings.

QUESTION NO: 37

To which switch or switches should you provide redundant links in order to achieve high availability with reliable fast convergence in the enterprise campus?

- A.** to a core switch running Cisco NSF and SSO from redundant distribution switches connected with a Layer 2 link
- B.** to a core switch running Cisco NSF and SSO from redundant distribution switches connected with a Layer 3 link
- C.** to two core switches from redundant distribution switches connected with Layer 2 link
- D.** to two core switches from redundant distribution switches connected with a Layer 3 link
- E.** to two core switches running Cisco NSF and SSO from two redundant distribution switches running Cisco NSF and SSO

Answer: D

Explanation:

QUESTION NO: 38

The requirement for high availability within the Data Center network may cause the designer to consider which one of the following solutions?

- A.** Construct a hierarchical network design using EtherChannel between a server and two VDCs from the same physical switch
- B.** Utilize Cisco NSF with SSO to provide intrachassis SSO at Layers 2 to 4
- C.** Define the Data Center as an OSPF NSSA area, advertising a default route into the DC And summarizing the routes out of the NSSA to the Campus Core
- D.** Implement network services for the Data Center as a separate services layer using active/active model that is more predictable in failure conditions

Answer: B

Explanation:

QUESTION NO: 39

Which four Cisco priority Spanning Tree Protocol enhancements are supported with rapid per-VLAN Spanning Tree? (Choose four)

- A. PortFast
- B. UplinkFast
- C. loop guard
- D. root guard
- E. BPDU guard
- F. BackboneFast

Answer: A,C,D,E

Explanation:

QUESTION NO: 40

When designing remote access to the Enterprise Campus network for teleworkers and mobile workers, which of the following should the designer consider?

- A. It is recommended to place the VPN termination device in line with the Enterprise Edge firewall, with ingress traffic limited to SSL only
- B. Maintaining access rules, based on the source IP of the client, on an internal firewall drawn from a headend RADIUS server is the most secure deployment
- C. VPN Headend routing using Reverse Route Injection (RRI) with distribution is recommended when the remote user community is small and dedicated DHCP scopes are in place
- D. Clientless SSL VPNs provide more granular access control than SSL VPN clients (thin or thick), including at Layer7

Answer: D

Explanation:

QUESTION NO: 41

Which EIGRP feature should a designer consider to limit the scope of EIGRP queries and minimize convergence time?

- A. Using multiple EIGRP processes
- B. Tuning down the EIGRP delay parameter
- C. EIGRP stub routing
- D. Limiting the number of EIGRP neighbor per device

Answer: C

Explanation:

QUESTION NO: 42

When designing the routing for an Enterprise Campus network it is important to keep which of the following route filtering aspects in mind?

- A. Filtering is only useful when combined with route summarization
- B. It is best to filter (allow) the default and summary prefixes only in the Enterprise Edge to remote sites or site-to-site IPsec VPN networks
- C. IGPs (for example EIGRP or OSPF) are superior to route filtering in avoiding inappropriate transit traffic through remote nodes or inaccurate or inappropriate routing updates
- D. The primary limitation of router filtering is that it can only be applied on outbound updates

Answer: B

Explanation:

QUESTION NO: 43

When considering the design of the IPv6 address plan for the Enterprise Campus network, which of the following should serve as guidance?

- A. All the IPv6 subnets should use a /32 prefix
- B. Set aside /31 prefixes to support point-to-point links and loopback interfaces
- C. The IPv6 address plan should be designed to support the service block model design or integration with IPv4
- D. Designate 16 subnet bits to be split up intelligently, either by OSPF area, VLAN numbering, or IPv4 mapping

Answer: D

Explanation:

QUESTION NO: 44

Which factor is least likely to affect the scalability of a VPN design?

- A. number of branch offices

- B. number of IGP routing peers
- C. remote Office and home worker throughput bandwidth requirements
- D. high availability requirements
- E. Supported applications

Answer: C

Explanation:

QUESTION NO: 45

Which of the following is true when considering the Server load-balancing design within the E-Commerce Module of the Enterprise Campus network?

- A. Routed mode requires the ACE run OSPF or EIGRP
- B. Bridged mode switches a packet between the public and the private subnets when it sees its MAC address as the destination
- C. Two-armed mode will place the SLB inline to the servers, with different client-side and a server-side VLANs
- D. One-armed mode, which uses the same VLAN for the client, the ACE, and the servers, requires a traffic-diversion mechanism to ensure the traffic return from the server passes through the ACE

Answer: D

Explanation:

QUESTION NO: 46

Which of the following is true regarding the effect of EIGRP queries on the network design?

- A. EIGRP queries will be the most significant issue with respect to stability and convergence
- B. EIGRP queries are not a consideration as long as EIGRP has a feasible successor with a next hop AD that is greater than the FD of the current successor route
- C. EIGRP queries will only increase the convergence time when there are no EIGRP stubs designed in the network

Answer: C

Explanation:

QUESTION NO: 47

Which two statements correctly identify considerations to take into account when deciding on Campus QoS Design elements? (Choose two)

- A. Voice needs to be assigned to the hardware priority queue
- B. Voice needs to be assigned to the software priority queue
- C. Call signaling must have guaranteed bandwidth service
- D. Strict-priority queuing should be limited to 50 percent of the capacity of the link
- E. At least 33 percent of the link bandwidth should be reserved for default best effort class

Answer: A,C

Explanation:

QUESTION NO: 48

Which version of spanning tree is recommended for the enterprise campus?

- A. CST
- B. MST
- C. STP
- D. PVST+
- E. PVRST+

Answer: E

Explanation:

QUESTION NO: 49

Which two design concerns must be addressed when designing a multicast implementation? (Choose two)

- A. only the low-order 23 bits of the MAC address are used to map IP addresses
- B. only the low-order 24 bits of the MAC address are used to map IP addresses
- C. only the high-order 23 bits of the MAC address are used to map IP address
- D. only the low-order 23 bits of the IP address are used to map MAC addresses
- E. the 0x01:00:00:00:00:00 MAC address prefix is used for mapping IP addresses to MAC addresses
- F. the 0x01:00:05:00:00:00 MAC address prefix is used for mapping IP addresses to MAC addresses

Answer: A,F

Explanation:

QUESTION NO: 50

What is the recommended practice regarding UDLD when implementing it in all fiber-optic LAN ports?

- A.** Adjust the default hello timers to three seconds for aggressive mode
- B.** Enable it in global mode and on every interface you need to support
- C.** Enable it in global mode to support every individual fiber-optic interface
- D.** Enable it to create channels containing up to eight parallel links between switches

Answer: C

Explanation:

QUESTION NO: 51

Which of the following two statements about Cisco NSF and SSO are the most relevant to the network designer? (Choose two)

- A.** You can reduce outages to 1 to 3 seconds by using SSO in a Layer 2 environment or Cisco NSF with SSO in a Layer 3 environment.
- B.** SSO and NSF each require the device to either be graceful restart-capable or graceful-aware.
- C.** In a fully redundant topology adding redundant supervisors with NSF and SSO may cause longer convergence times than single supervisors with tuned IGP timers
- D.** The primary deployment scenario for Cisco NSF with SSO is in the Distribution and Core layers.
- E.** Cisco NSF-aware neighbor relationship are independent of any turned IGP times

Answer: A,C

Explanation:

QUESTION NO: 52

Which of these statements about FSPF is true?

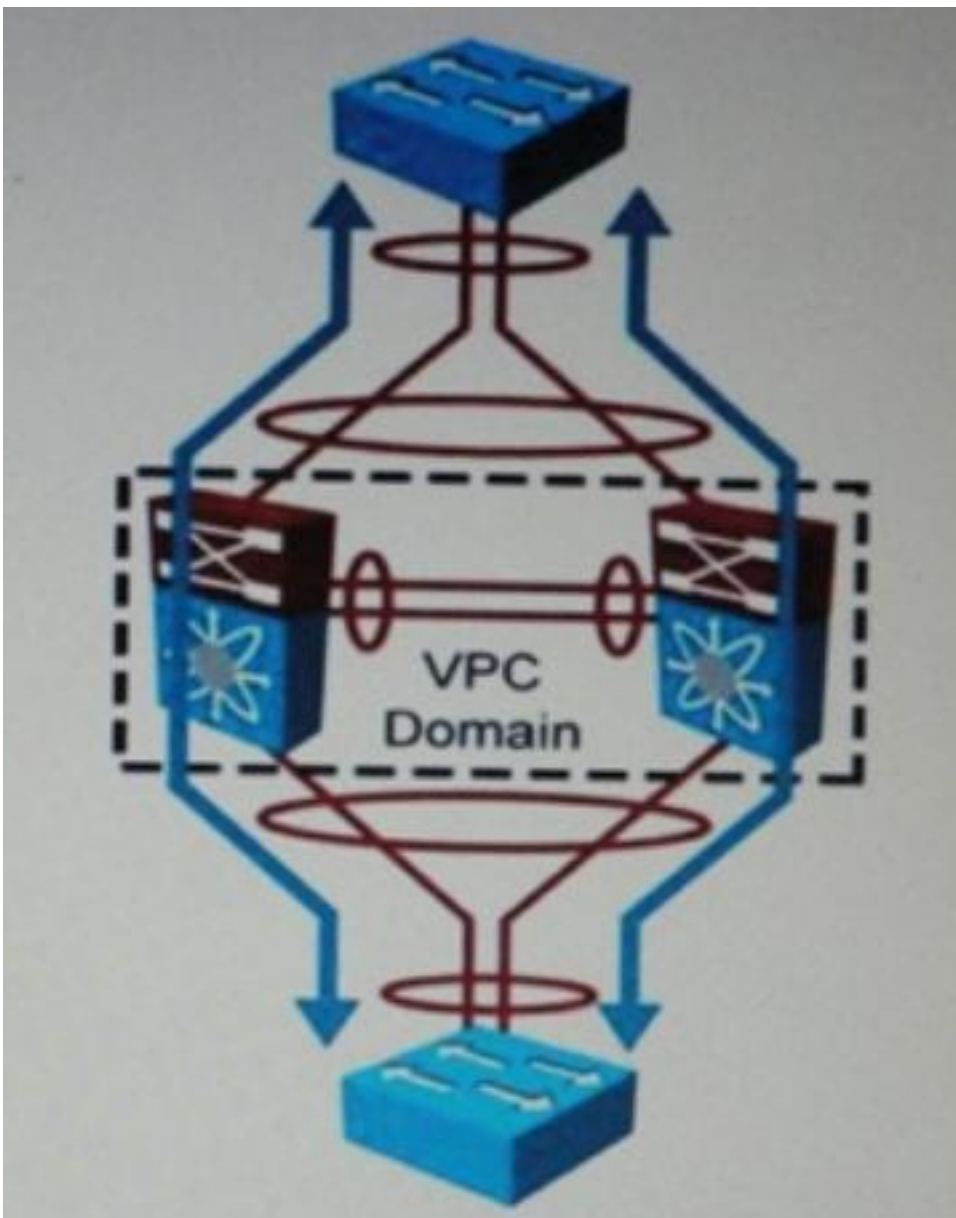
- A. It supports multipath routing
- B. It can run any type of storage ports
- C. When it is used, hop-by-hop routes are based only on the switch ID
- D. When it is used, path status is based on the functionality of attached ports
- E. It runs only on a switch fabric and cannot function in a VSAN

Answer: A

Explanation:

QUESTION NO: 53

Refer to the exhibit



Which of the following is an advantage of device clustering utilizing Virtual Port Channels (vPC)?

- A. A logical star topology provides a loop free environment so that all links will be used to forward traffic
- B. Enhanced EtherChannel hashing load balancing using the vPC peer link internal to the VPC
- C. The control plane functions of the Nexus switches are merged to hide the use of virtualization
- D. Neighboring devices connect on a Layer 3 MEC for improved packet forwarding

Answer: A

Explanation:

QUESTION NO: 54

Which three statements about firewall modes are correct? (Choose three)

- A. A firewall in routed mode has one IP address
- B. A firewall in transparent mode has one IP address
- C. In routed mode, the firewall is considered to be a Layer 2 device
- D. In routed mode, the firewall is considered to be a Layer 3 device
- E. In transparent mode, the firewall is considered to be a Layer 2 device
- F. In transparent mode, the firewall is considered to be a Layer 3 device

Answer: B,D,E

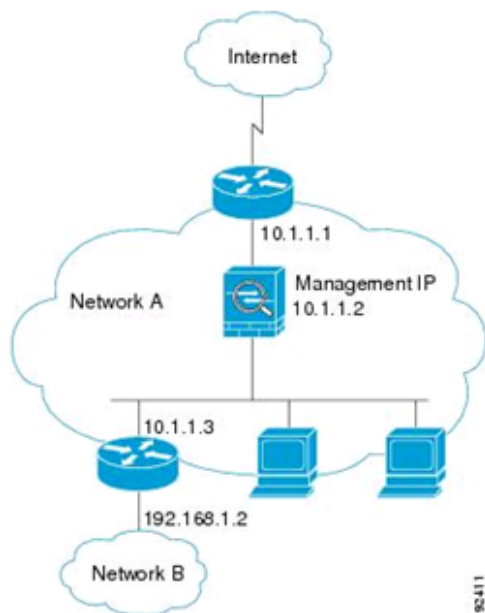
Explanation:

In "Designing Cisco Network Service Architectures (ARCH)" it is clearly stated on page 334:

"A transparent firewall has one IP address assigned to the entire bridge group, and uses this management address as the source address for packets originated on the firewall."

Incorrect answer:

„In transparent mode, the firewall is considered to be a Layer 3 device“ is incorrect:



Transparent Firewall Guidelines

Follow these guidelines when planning your transparent firewall network:

- A management IP address is required; for multiple context mode, an IP address is required for each context.

QUESTION NO: 55

Which one of these statements is true concerning the data center distribution (aggregation) layer design?

- A.** With Layer 3 at the aggregation layer, the physical loops in the topology must still be managed by STP.
- B.** The boundary between Layer 2 and Layer 3 must reside in the multilayer switches, independent of any other devices such as firewalls or content switching devices.
- C.** A mix of both Layer 2 and Layer 3 access is sometimes the most optimal.
- D.** In a small data center, the aggregation layer can connect directly to the campus core, exchanging IP routes and MAC address tables.

Answer: B

Explanation:

QUESTION NO: 56

Which unique characteristics of the Data Center Aggregation layer must be considered by an Enterprise Campus designer?

- A.** Layer 3 routing between the Access and Aggregation layers facilitates the ability to span VLANs across multiple access switches, which is a requirement for many server virtualization and clustering technologies.
- B.** "East-west" server-to-server traffic can travel between aggregation modules by way of the core, but backup and replication traffic typically remains within an aggregation module.
- C.** Load balancing, firewall services, and other network services are commonly integrated by the use of service modules that are inserted in the aggregation switches.
- D.** Virtualization tools allow a cost effective approach for redundancy in the network design by using two or four VDCs from the same physical switch.

Answer: C

Explanation:

QUESTION NO: 57

Which of the following two statements about Cisco NSF and SSO are the most relevant to the network designer? (Choose two)

- A.** You can reduce outages to 1 to 3 seconds by using SSO in a Layer 2 environment or Cisco NSF with SSO in a Layer 3 environment.
- B.** SSO and NSF each require the device to either be graceful restart-capable or graceful-aware.
- C.** In a fully redundant topology adding redundant supervisors with NSF and SSO may cause longer convergence times than single supervisors with tuned IGP timers
- D.** The primary deployment scenario for Cisco NSF with SSO is in the Distribution and Core layers.
- E.** Cisco NSF-aware neighbor relationship are independent of any turned IGP times

Answer: A,C

Explanation:

QUESTION NO: 58

Which four Cisco proprietary Spanning Tree Protocol enhancements are supported with rapid per-VLAN Spanning-Tree plus? (Choose four.)

- A.** PortFast

- B. UnlinkFast
- C. loop guard
- D. root guard
- E. BPDU guard
- F. BackboneFast

Answer: A,C,D,E

Explanation:

QUESTION NO: 59

You are the Cisco Network Designer in Cisco.com. Your company is using the G.711 codec with 802.11a access point radios. This can support a maximum of how many phones per access point?

- A. 5
- B. 10
- C. 14
- D. 20

Answer: C

Explanation:

In "Designing Cisco Network Service Architectures (ARCH)", page 512

It says: "In comparison, 802.11a AP radios can support 14 active voice calls using the G.711 codec".

QUESTION NO: 60

With respect to address summarization, which of the following statements concerning IPv4 and IPv6 is true?

- A. The potential size of the IPv6 address blocks suggests that address summarization favors IPv6 over IPv4.
- B. Role based addressing using wildcard masks to match multiple subnets is suitable for IPv4, but unsuitable for IPv6.
- C. In order to summarize, the number of subnets in the IPv4 address block should be a power of 2 while the number of subnets in the IPv6 address block should be a power of 64.
- D. WAN link addressing best supports summarization with a/126 subnet for IPv4 and a/31 for IPv6.

Answer: B

Explanation:

QUESTION NO: 61

The Cisco Nexus 1000V is intended to address which disadvantage of the VMware vSphere solution?

- A.** Inability to deploy new functional servers without requiring physical changes on the network
- B.** Complexity added by the requirement for an ESX host for each virtual machine
- C.** Network administrators lack control of the access layer of the network
- D.** To increase the number of physical infrastructure and the virtual machines that can be managed

Answer: C

Explanation:

QUESTION NO: 62

Which of the following facts must be considered when designing for IP telephony within an Enterprise Campus network?

- A.** Because the IP phone is a three-port switch. IP telephony extends the network edge, impacting the Distribution layer.
- B.** Video and voice are alike in being bursty and bandwidth intensive, and thus impose requirements to be lossless, and have minimized and jitter.
- C.** IP phones have no voice and data VLAN separation, so security policies must be based on upper layer traffic characteristics.
- D.** Though multi-VLAN access ports are set to dot1q and carry more than two VLANs they are not trunk ports.

Answer: D

Explanation:

QUESTION NO: 63

Support of vPC on the Cisco Nexus 5000 access switch enables various new design options for the data center Access layer, including which of the following?

- A.** The vPC peer link is not required for Access layer control traffic, and can instead be used to span VLANs across the vPC access switches
- B.** A single switch can associate per-interface with more than one vPC domain
- C.** vPC can be used on both sides of the MEC, allowing a unique 16-link EtherChannel to be built between the access and aggregation switches
- D.** Allows an EtherChannel between a server and a access switch while still maintaining the level of availability that is associated with dual-homing a server to two different access switches

Answer: C

Explanation:

QUESTION NO: 64 DRAG DROP

Drag the OSPF technology on the left to the appropriate network convergence step on the right that this technnnology helps to mitigat



Answer:



Explanation:

BFD – Detect the event
 LSA Propagation – Propagate the event
 SPF Throttling – Process the event
 CEF – Update forward data structure

QUESTION NO: 65

Which of these recommendations is most appropriate for the Cisco Campus Architecture?

- A. Utilize Layer 3 switching.
- B. Utilize software accelerated services.
- C. Aggregate end users and support a feature-rich environment.
- D. Perform packet manipulation and filtering at the core layer.
- E. Use redundant point-to-point Layer 2 interconnections when there is a link or node failure.

Answer: A

Explanation:

QUESTION NO: 66

Which two restrictions must the Enterprise Campus network designer consider when evaluating WAN connectivity options? (Choose two.)

- A. OSPF over a multiaccess EMS or VPLS network may not have consistent broadcast or multicast performance
- B. IP multicast is not supported over Layer 3 MPLS VPN; instead a Layer 2 MPLS VPN must be utilized with service provider support
- C. QoS requirements with MPLS-VPNs must be implemented by the service provider
- D. Hierarchical VPLS designs are the least scalable
- E. IGMP snooping is not an option with VPLS or EMS; instead administrative scoping or allowing sufficient bandwidth for unnecessary multicast traffic at the edge links is required

Answer: A,E

Explanation:

QUESTION NO: 67

Which Virtualization technology does not need to enforce separation of the control plane?

- A. Server Virtualization using vSphere
- B. Network virtualization using VRFs
- C. Device clustering using VSS
- D. Device virtualization using VMWare

Answer: D

Explanation:

QUESTION NO: 68

Which protocol will not adhere to the design requirement of the control plane being either separated or combined within a virtualization technology?

- A. FHRP
- B. STP
- C. CEF
- D. NSF with SSO

Answer: B

Explanation:

QUESTION NO: 69

Which of the following two are advantages of Server virtualization using VMware vSphere?
(Choose two)

- A. Retains the one-to-one relationship between switch ports and functional servers
- B. Enables the live migration of a virtual server from one physical server to another without disruption to users or loss of services
- C. The access layer of the network moves into the vSphere ESX servers, providing streamlined vSphere management
- D. Provides management functions including the ability to run scripts and to install third-party agents for hardware monitoring, backup, or systems management
- E. New functional servers can be deployed with minimal physical changes on the network

Answer: B,D

Explanation:

QUESTION NO: 70

Addressing QoS design in the Enterprise Campus network for IP Telephony applications means what?

- A.** It is critical to identify aggregation and rate transition points in the network, where preferred traffic and congestion QoS policies should be enforced
- B.** Suspect traffic should be dropped closest to the source, to minimize wasting network resources
- C.** An Edge traffic classification scheme should be mapped to the downstream queue configuration
- D.** Applications and Traffic flows should be classified, marked and policed within the Enterprise Edge of the Enterprise Campus network

Answer: A

Explanation:

QUESTION NO: 71

A well-designed IP addressing scheme supporting role-based functions within the subnet will result in the most efficient use of which technology?

- A.** Layer 3 switching in the core
- B.** Network Admission Control (NAC)
- C.** IP telephony (voice and video) services
- D.** ACLs

Answer: D

Explanation:

QUESTION NO: 72

Which of the following features might be used by the Enterprise Campus network designer as a means of route filtering?

- A.** IPv4 static routes
- B.** Route tagging using a route map in an ACL
- C.** Tagging routes using the BGP MED
- D.** EIGRP stub networks

Answer: D

Explanation:

QUESTION NO: 73

Which of the following is a result when designing multiple EIGRP autonomous systems within the Enterprise Campus network?

- A. Improves scalability by dividing the network using summary routes at AS boundaries
- B. Decreases complexity since EIGRP redistribution is automatically handled in the background
- C. Reduces the volume of EIGRP queries by limiting them to one EIGRP AS
- D. Scaling is improved when a unique AS is run at the Access, Distribution, and Core layers of the network

Answer: A

Explanation:

QUESTION NO: 74

Which of the following two are effective and simple means of employing route summarization within the Enterprise Campus network? (Choose two)

- A. A default route (0.0.0.0 /0) advertised dynamically into the rest of the network
- B. Route filtering to manage traffic flows in the network, avoid inappropriate transit traffic through remote nodes, and provide a defense against inaccurate or inappropriate routing updates
- C. Use manual split horizon
- D. Use a structured hierarchical topology to control the propagation of EIGRP queries
- E. Open Shortest Path First (OSPF) stub areas

Answer: A,E

Explanation:

QUESTION NO: 75

The network designer needs to consider the number of multicast applications and sources in the network to provide the most robust network possible. Which of the following is a consideration the designer must also address?

- A. The IGPs should utilize authentication to avoid being the most vulnerable component
- B. With SSM source or receiver attacks are not possible
- C. With Shared Trees access control is always applied at the RP
- D. Limit the rate of Register messages to the RP to prevent specific hosts from being attacked on a PIM-SM network

Answer: B

Explanation:

QUESTION NO: 76

Which statement is the most accurate regarding IPsec VPN design for an Enterprise Campus environment?

- A.** VPN device IP addressing must align with the existing Campus addressing scheme.
- B.** The choice of a hub-and-spoke or meshed topology ultimately depends on the number of remotes.
- C.** Sizing and selection of the IPsec VPN headend devices is most affected by the throughput bandwidth requirements for the remote offices and home worker
- D.** Scaling considerations such as headend configuration, routing protocol choice, and topology have the broadest impact on the design.

Answer: D

Explanation:

QUESTION NO: 77

When considering the design of the E-Commerce topology which of the following are true?

- A.** One-armed SLB design with multiple security contexts removes the need for a separate firewall in the core layer
- B.** Two-firewall-layer SLB design considers the aggregation and access layers to be trusted zones, requiring no security between the web, application, and database zones
- C.** One-armed SLB design with two firewall layers ensures that non load-balanced traffic still traverses the ACE so that the health and performance of the servers is still being monitored
- D.** In all cases the will be configuration requirements for direct access to any servers or for nonload-balanced sessions initiated by the servers

Answer: A

Explanation:

QUESTION NO: 78

Distinct, physical redundancy within a network layer is a key characteristic that contributes to the high availability of the hierarchical network design. Which of the following is not an examples of

this model?

- A. SAN extension with dual fabrics such as a yellow VSAN and a blue VSAN utilized via multipath software
- B. Redundant power supplies and hot-swappable fan trays in Aggregate switches
- C. A single SAN fabric with redundant uplinks and switches
- D. Servers using network adapter teaming software connected to dual-attached access switches

Answer: C

Explanation:

QUESTION NO: 79

Which of the following is most accurate with respect to designing high availability within the Enterprise Campus network?

- A. High availability at and between the Distribution and Access layers is as simple as redundant switches and redundant Layer 3 connections
- B. Non-deterministic traffic patterns require a highly available modular topology design
- C. Distribution layer high availability design includes redundant switches and Layer 3 equal-cost load sharing connections to the switched Access and routed Core layers, with a Layer 3 link between the Distribution switches to support summarization of routing information from the Distribution to the Core
- D. Default gateway redundancy allows for the failure of a redundant Distribution switch without affecting endpoint connectivity

Answer: D

Explanation:

QUESTION NO: 80

Which of the following should the Enterprise Campus network designer consider with respect to Video traffic?

- A. While it is expected that the sum of all forms of video traffic will grow to over 90% by 2013, the Enterprise will be spared this rapid adoption of video by consumers through a traditional top-down approach
- B. Avoid bandwidth starvation due to video traffic by preventing and controlling the wide adoption of unsupported video applications
- C. Which traffic model is in use, the flow direction for the traffic streams between the application

components, and the traffic trends for each video application

D. Streaming video applications are sensitive to delay while interactive video applications, using TCP as the underlying transport, are fairly tolerant of delay and jitter

Answer: C

Explanation:

QUESTION NO: 81

Which two protocol characteristics should be most considered when designing a single unified fabric for the Data Center? (Choose two.)

A. FCIP or FCoE allow for easier integration by using the Fibre Channel Protocol (FCP) and Fibre Channel framing

B. iSCSI uses a special EtherType and an additional header containing additional control information

C. FCIP and iSCSI has higher overhead than FCoE owing to TCP/IP

D. FCoE was initially developed to be used as a switch-to-switch protocol, while FSIP is primarily meant to be used as an access layer protocol to connect hosts and storage to a Fibre Channel SAN

E. FCoE requires gateway functionality to integrate into an existing Fibre Channel network

Answer: A,C

Explanation:

Topic 2, Volume B

QUESTION NO: 82

Cisco Express Forwarding (CEF) is mainly used to increase packet switching speed, reducing the overhead and delays introduced by other routing techniques, increasing overall performance.

Which of the following concerning CEF is recommended by Cisco?

A. Use default Layer 4 hash in core.

B. Use default Layer 3 hash in distribution.

C. Use default Layer 4 hash in distribution.

D. Use default Layer 3 hash in core and Layer 3 + Layer 4 hash in distribution layer.

Answer: D

Explanation:

QUESTION NO: 83

Which typical enterprise campus requirement ensures that the network supports the required applications and that data flows within the required time frames?

- A. availability
- B. performance
- C. functionality
- D. manageability

Answer: C

Explanation:

QUESTION NO: 84

You are the Cisco Network Designer in Cisco.com. Which of these is a Layer 2 transport architecture that provides packet-based transmission optimized for data based on a dual ring topology?

- A. Dynamic Trunking Protocol
- B. Resilient Packet Ring
- C. Synchronous Digital Hierarchy
- D. Coarse Wave Division Multiplexing

Answer: B

Explanation:

QUESTION NO: 85

What two choices can you make when redundancy is required from a branch office to a regional office? (Choose two.)

- A. multiple Frame Relay PVCs
- B. dual Wan links to the regional office
- C. dual Wan links to another branch office
- D. single links - one to the regional office and one to another branch office

Answer: B,D

Explanation:

QUESTION NO: 86

Which one is not the feature of the Cisco Unified Wireless Network architecture?

- A. network unification
- B. remote access
- C. mobility services
- D. network management

Answer: B

Explanation:

QUESTION NO: 87

What type of Call Admission control in CallManager allows for limits to the bandwidth consumed by active calls?

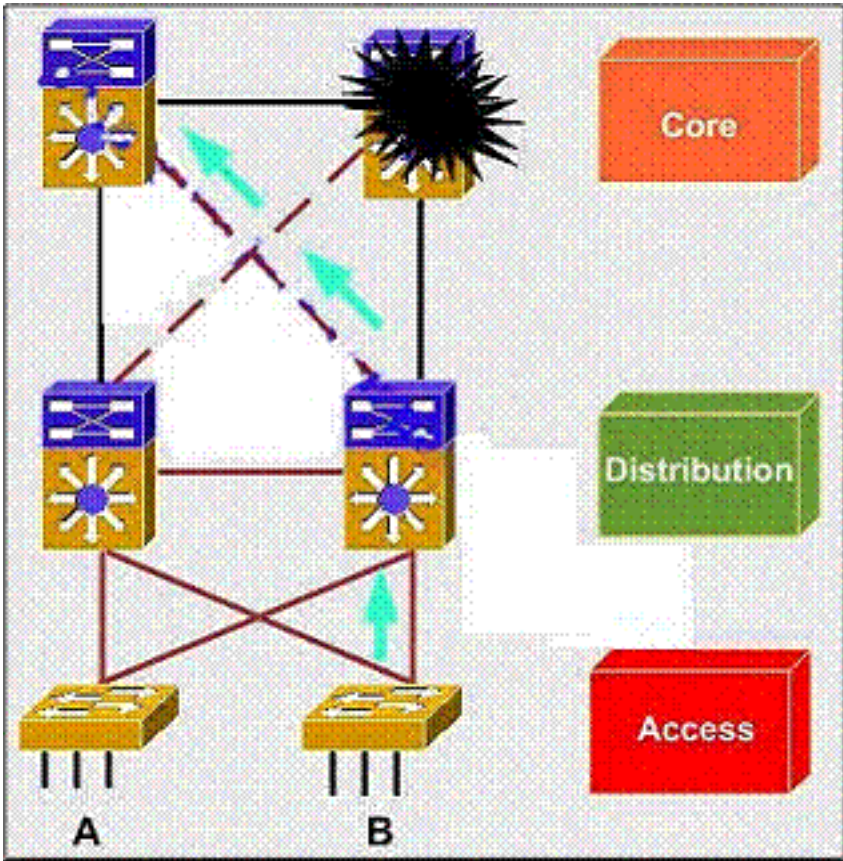
- A. regions
- B. partitions
- C. locations
- D. device Pools

Answer: C

Explanation:

QUESTION NO: 88

Which two of these are correct regarding the recommended practice for distribution layer design based on the following configuration?



- A. use a Layer 2 link between distribution switches
- B. use a Layer 3 link between distribution switches
- C. use a redundant link to the core
- D. use a Layer 3 link between distribution switches with route summarization

Answer: C,D

Explanation:

QUESTION NO: 89

Which VPN management feature would be considered to ensure that the network had the least disruption of service when making topology changes?

- A. dynamic reconfiguration
- B. path MTU discovery
- C. auto setup
- D. remote management

Answer: A

Explanation:

Dynamic reconfiguration: All configuration changes should take effect without requiring a reboot of the device. Disruption of service with a fully loaded VPN device can potentially impact thousands

of individual users.

Reference: Arch student guide p.9-17

QUESTION NO: 90

Jitter is an unwanted variation of one or more characteristics of a periodic signal in electronics and telecommunications and _____ refers to call issues that cause variations in timing or time of arrival

- A. echo
- B. jitter
- C. packet loss
- D. digitized sampling

Answer: B

Explanation:

QUESTION NO: 91

Which three components are part of the Intelligent Network Services provided by the Cisco AVVID framework? (Choose three.)

- A. IP telephony
- B. security
- C. IP multicasting
- D. QoS

Answer: B,C,D

Explanation:

QUESTION NO: 92

Cisco IDS sensors form the eyes and ears of your Cisco network intrusion detection system. Placing sensors correctly throughout your network is crucial to successfully implementing your Cisco intrusion detection system .Which two of these are characteristics of an IDS sensor? (Choose two.)

- A. has a permissive interface that is used to monitor networks

- B. is an active device in the traffic path
- C. passively listens to network traffic
- D. has a promiscuous interface that is used to monitor the network

Answer: C,D

Explanation:

QUESTION NO: 93

Which three best practices should be implemented at the campus backbone submodule to support the server farm module? (Choose three.)

- A. Implement highly redundant switching and links with no single points or paths of failure.
- B. Implement server load balancing.
- C. Implement the Hot Standby Router Protocol (HSRP) for failover protection.
- D. Implement intrusion detection with automatic notification of intrusion attempts in place.

Answer: A,C,D

Explanation:

QUESTION NO: 94

As an experienced technician, you are responsible for Technical Support. One of the trainees is asking your advice on VPN Termination Device and Firewall Placement. Which of the following approaches will you recommend?

- A. inline with a firewall
- B. in a DMZ outside the firewall
- C. parallel with a firewall
- D. in a DMZ behind the firewall

Answer: D

Explanation:

QUESTION NO: 95

The network administrator would like to generate synthetic traffic using the Service Assurance Agent contained in Cisco IOS. Which CiscoWorks network management application will be used to report the latency and availability for configured traffic operations on an end-to-end and hop-by-hop (router-to router) basis?

- A. nGenius Real-Time Monitor
- B. CiscoView
- C. Device Fault Manager
- D. Internetwork Performance Monitor

Answer: D

Explanation:

QUESTION NO: 96

Cisco IDS sensors form the eyes and ears of your Cisco network intrusion detection system. Placing sensors correctly throughout your network is crucial to successfully implementing your Cisco intrusion detection system . Where can an IPS sensor be placed in an enterprise network? (Choose two.)

- A. core layer
- B. bridging two VLANs on one switch
- C. between two Layer 2 devices with trunking
- D. between two Layer 2 devices without trunking

Answer: C,D

Explanation:

QUESTION NO: 97

Which protocol would provide block access to remote storage over WAN links?

- A. iSCSI
- B. FCIP
- C. SCSI-FP
- D. eSCSI

Answer: A

Explanation:

QUESTION NO: 98

The Cisco network-based virtual firewall service solution helps service providers to deliver cost-effective, scalable, integrated security services for enterprise customers using Cisco platforms .What is a virtual firewall?

- A. another name for a firewall deployed in routed mode
- B. another name for a firewall deployed in transparent mode
- C. a separation of multiple firewall security contexts on a single firewall
- D. a firewall that, when deployed in routed mode, can support up to 1000 VLANs per context

Answer: C

Explanation:

QUESTION NO: 99

What is the device weight limit per CallManager in a Cisco IP phone configuration?

- A. 2500
- B. 3000
- C. 5000
- D. 6500

Answer: C

Explanation:

QUESTION NO: 100

In a VoWLAN deployment, It is recommended ____ dBm separation between cells with the same channel.

- A. 6
- B. 7
- C. 10
- D. 19

Answer: D

Explanation:

QUESTION NO: 101

Acme Nutrition manufactures a wide variety of vitamin supplements. It has a single manufacturing facility with 3 regional warehouses and 16 district sales offices. Currently the manufacturing facility requires 210 IP addresses; each warehouse requires 51 IP addresses; each district sales office requires 11 IP addresses; and the IP WAN requires 38 IP addresses. if Acme Nutrition plans for 20 percent growth in facilities, how many Class C subnets will the district sales offices require?

- A. 19 (3 from the warehouse range and 16 from a separate Class C address)
- B. 19 (3 from the warehouse block, 15 from a separate Class C block and 1 from the IP WAN block)
- C. 20 (4 from the warehouse range, 15 from a separate Class C block and 1 from the IP WAN block)
- D. 16 (3 from the warehouse range and 13 from a separate Class C address)

Answer: B

Explanation:

QUESTION NO: 102

You are the Cisco Network Designer. Which of these is least important when determining how many users a NAS can support?

- A. bandwidth
- B. number of plug-ins per scan
- C. total number of network devices
- D. number of checks in each posture assessment

Answer: A

Explanation:

QUESTION NO: 103

When designing the WAN module within the enterprise edge, which document is used to specify the connectivity and performance agreements with the service provider?

- A. RFP
- B. RFC
- C. SLC/SLA
- D. SOW

Answer: C

Explanation:

QUESTION NO: 104

Which site-to-site VPN solution allows Cisco routers, PIX Firewalls, and Cisco hardware clients to act as remote VPN clients in order to receive predefined security policies and configuration parameters from the VPN headend at the central site?

- A. Easy VPN
- B. GRE tunneling
- C. Virtual Tunnel Interfaces
- D. Dynamic Multipoint VPN
- E. Group Encrypted Transport VPN

Answer: A

Explanation:

QUESTION NO: 105

Which routing protocol supports a flexible area structure using routing levels one and two?

- A. OSPF
- B. EIGRP
- C. IS-IS
- D. BGP

Answer: C

Explanation:

QUESTION NO: 106

Please match the Cisco STP enhancement term to its definition.(Not all options will be used.)

(1) BPDU guard

(2) PortFast

(3) BackboneFast

(4) UplinkFast

(5) Loop guard

(a) shuts down a port that receives a BPDU when enabled

(b) cuts convergence time by max-age for indirect failure

(c) prevents the alternate or root port from being designated in absence of BPDUs

(d) causes Layer 2 LAN interface access port to immediately enter the forwarding state

(f) helps prevent bridging loops due to uni-directional link failures on point-to-point links

A. (a)-(1);(b)-(2);(c)-(4);(d)-(5);(e)-(3)

B. (a)-(4);(b)-(3);(c)-(2);(d)-(5);(e)-(1)

C. (a)-(3);(b)-(2);(c)-(4);(d)-(5);(e)-(1)

D. (a)-(1);(b)-(4);(c)-(3);(d)-(5);(e)-(2)

Answer: D

Explanation:

Cisco offers a variety of enhancements to STP:

1. PortFast: Allows an access port to bypass STPs listening and learning phases so no need to wait 50 seconds to forward data.

2. UplinkFast: Reduces STP convergence from 50 seconds to approximately 3 to 5 seconds so no need to wait 50 seconds to forward data through alternate link

3. BackboneFast: Reduces STP convergence time for an indirect link failure.

4. LoopGuard: Helps prevent loops that could occur because of a unidirectional link failure, a software failure, or a bridge protocol data unit (BPDU) loss due to congestion

5. RootGuard: Prevents an inappropriate switch from being elected as a root bridge

6. BPDUGuard: Causes a port configured for PortFast to go into the errordisable state if a BPDU is received on the port

QUESTION NO: 107

When is the site-to-site remote access model appropriate? (Choose one.)

- A. for multiple ISDN connections
- B. for modem concentrated dial-up connections
- C. for a group of users in the same vicinity sharing a connection
- D. for use by mobile users

Answer: C

Explanation:

QUESTION NO: 108

Which two of these are recommended practices with trunks? (Choose two.)

- A. use ISL encapsulation
- B. use 802.1q encapsulation
- C. set ISL to desirable and auto with encapsulation negotiate to support ILS protocol negotiation
- D. use VTP server mode to support dynamic propagation of VLAN information across the network
- E. set DTP to desirable and desirable with encapsulation negotiate to support DTP protocol negotiation.

Answer: B,E

Explanation:

QUESTION NO: 109

What are three primary activities in the cycle of building an enterprise security strategy? (Choose three.)

- A. activity audit
- B. administration
- C. policy establishment
- D. technology implementation

Answer: A,C,D

Explanation:

QUESTION NO: 110

For acceptable voice calls, the packet error rate should be less than ____%

- A. 0.01
- B. 0.1
- C. 1
- D. 2.5

Answer: C

Explanation:

QUESTION NO: 111

What are two design guidelines for VoIP networks? (Choose two.)

- A. Delay should be no more than 10 ms.
- B. Loss should be no more than 1 percent.
- C. Jitter should be less than 40 ms.
- D. Managed bandwidth is strongly recommended for voice control traffic.

Answer: B,D

Explanation:

QUESTION NO: 112

You are the Cisco Network Designer in Cisco.com. Which of these statements is true of clientless end-user devices?

- A. They do not receive unique IP addresses.
- B. RADIUS or LDAP is required.
- C. They are assigned addresses from the internal DHCP pool.
- D. Their traffic appears to originate from the originating host network.

Answer: A

Explanation:

QUESTION NO: 113

Users of a site-to-site VPN are reporting performance problems. The VPN connection employs IPSec and GRE and traverses several Ethernet segments. The VPN packets are being fragmented as they traverse the links. What would be two methods to overcome this problem? (Choose two.)

- A. Employ path MTU discovery.
- B. Set the MTU higher than 1500 bytes.
- C. Turn off pre-fragmentation for IPSec.
- D. Set the MTU value to 1400 bytes.

Answer: A,D

Explanation:

QUESTION NO: 114

You are the Cisco Network Designer in Cisco.com. Which statement is correct regarding NBAR and NetFlow?

- A. NBAR examines data in Layers 1 and 4.
- B. NBAR examines data in Layers 3 and 4.
- C. NetFlow examines data in Layers 3 and 4.
- D. NBAR examines data in Layers 2 through 4.

Answer: B

Explanation:

QUESTION NO: 115

Lafayette Productions is looking for a new ISP that has improved availability, load balancing, and catastrophe protection. Which type of ISP connectivity solution would be best?

- A. single run
- B. multi-homed
- C. stub domain EBGP
- D. direct BGP peering

Answer: B

Explanation:

QUESTION NO: 116

It's a configuration that experts are calling a "firewall sandwich," with the second firewall providing a second level of load balancing after traffic down. What is meant by the term "firewall sandwich"?

- A. single layer of firewalling
- B. multiple layers of firewalling
- C. firewall connections in either an active or standby state
- D. an architecture in which all traffic between firewalls goes through application-specific servers

Answer: B

Explanation:

QUESTION NO: 117

To securely transport EIGRP traffic, a network administrator will build VPNs between sites. What is the best method to accomplish the transport of EIGRP traffic?

- A. IPsec in tunnel mode
- B. IPsec in transport mode
- C. GRE with IPsec in transport mode
- D. GRE with IPsec in tunnel mode

Answer: D

Explanation:

The right answer is "GRE with IPsec in tunnel mode".

In "Designing Cisco Network Service Architectures (ARCH)" is explains, that in order to transport EIGRP routing updates, it is GRE over IPsec tunnel. See page 404.

QUESTION NO: 118

When BGP tuning is used, how is packet flow into the e-commerce module controlled?

- A. by tracking the status of objects along the path to the e-commerce module
- B. by detecting undesirable conditions along the path to the e-commerce module
- C. by using the MED to communicate the site preferences for traffic to multiple ISPs
- D. by communicating the available prefixes, routing policies, and preferences of each site to its ISP
- E. by moving the SLB to a position where selected traffic to and from the servers does not go through the SLB

Answer: D

Explanation:

QUESTION NO: 119

Which three objectives would be met by designing Layer 3 switching in the Campus Backbone of a medium size installation? (Choose three.)

- A. scale to a large size
- B. increase router peering
- C. provide a flexible topology with no spanning tree loops
- D. control broadcasts in the backbone

Answer: A,C,D

Explanation:

QUESTION NO: 120

You are the Cisco Network Designer. Which is not major scaling, sizing, and performance consideration for an IPsec design?

- A. connection speed
- B. number of remote sites
- C. features to be supported
- D. types of devices at the remote site

Answer: A

Explanation:

QUESTION NO: 121

Which enterprise caching mode eliminates the need for Layer 4 switches or WCCP enabled routers to intercept user requests?

- A. transparent
- B. proxy
- C. reverse proxy
- D. direct

Answer: B

Explanation:

In proxy mode, end-user web browsers need to be explicitly configured to the IP address or host name of the Content Engine, and there is no need for additional hardware such as Layer 4 switches or Web Cache Communication Protocol (WCCP)-enabled routers to intercept user requests, as in transparent caching. Enterprises are normally interested in deploying transparent network caching, but some enterprises may have a legacy requirement for a proxy (nontransparent) cache.

Reference: Arch student guide p.12-12

QUESTION NO: 122

Which signal and noise values will result in the best phone communication with an access point?

- A. signal strength 46dBm, noise level 95dBm
- B. signal strength 74dBm, noise level 94dBm
- C. signal strength 68dBm, noise level 79dBm
- D. signal strength 50dBm, noise level 56dBm

Answer: A

Explanation:

QUESTION NO: 123

What are two considerations to using IP Multicast delivery? (Choose two.)

- A. no congestion avoidance

- B. not for bandwidth intensive applications
- C. no guaranteed delivery mechanism
- D. source sends multiple data streams out each interface

Answer: A,C

Explanation:

QUESTION NO: 124

Which remote access VPN addressing technique supports a static IP address to support a specific application?

- A. Use a static ip addresses based on incoming user policies.
- B. Use DHCP to assign addresses based on incoming user policies.
- C. Deploy a clientless model to assign a unique address to the user.
- D. Deploy RADIUS or LDAP to assign the address to the user.

Answer: D

Explanation:

QUESTION NO: 125

Which three are used in configuring Call Manager dial plans? (Choose three.)

- A. route list
- B. route group
- C. gateway list
- D. route pattern

Answer: A,B,D

Explanation:

QUESTION NO: 126

Which two of these are characteristics of an IPS device? (Choose two.)

- A. passively listens to network traffic
- B. is an active device in the traffic path
- C. has a permissive interface that is used to monitor networks
- D. traffic arrives on one IPS interface and exits on another

Answer: B,D

Explanation:

QUESTION NO: 127

Which three LAN routing protocols would be appropriate for a small retail organization with a multi-vendor LAN infrastructure? (Choose three.)

- A. IGRP
- B. RIP
- C. RIPv2
- D. OSPF

Answer: B,C,D

Explanation:

QUESTION NO: 128

One of your customer has six sites, three of which process a large amount of traffic among them. He plans to grow the number of sites in the future. Which is the most appropriate design topology?

- A. full mesh
- B. peer-to-peer
- C. partial mesh
- D. hub and spoke

Answer: C

Explanation:

QUESTION NO: 129

ABC Company has 1500 managed devices and 15,000 end users on a campus network. LAN Management Solution (LMS) is being deployed as the network management application. What is the recommended number of network management server(s)?

- A. 1
- B. 2
- C. 3
- D. 4

Answer: A

Explanation:

QUESTION NO: 130

You are the network consultant from Cisco.com. Your customer has eight sites and will add in the future.

Branch site to branch site traffic is approaching 30 percent. The customer's goals are to make it easier to add branch sites in the future and to reduce traffic through the hub. Which VPN topology should you recommend?

- A. Easy VPN
- B. IPsec GRE tunneling
- C. Virtual Tunnel Interfaces
- D. Dynamic Multipoint VPN

Answer: D

Explanation:

QUESTION NO: 131

The Schuyler and Livingston Iron Works has been working on getting its network security under control. It has set up VPN with IPSec links to its suppliers. It has installed network vulnerability scanners to proactively identify areas of weakness, and it monitors and responds to security events as they occur. It also employs extensive access control lists, stateful firewall implementations, and dedicated firewall appliances. The company has been growing very fast lately and wants to make sure it is up to date on security measures. Which two areas of security would you advise the company to strengthen? (Choose two.)

- A. intrusion protection
- B. identity
- C. secure connectivity
- D. security management

Answer: A,B

Explanation: The right answer should be identity and intrusion protection (A,B) because security management is covered by the vulnerability scanner and monitor.

Topic 3, Volume C

QUESTION NO: 132

Which two of these key fields are used to identify a flow in a traditional NetFlow implementation? (Choose two.)

- A. source port
- B. output interface
- C. next-hop IP address
- D. source MAC address
- E. destination IP address
- F. next-hop MAC address

Answer: A,E

Explanation:

QUESTION NO: 133

Users at the Charleville Company began experiencing high network delays when Internet connectivity was enabled for all users. After investigating the traffic flow, you determine that peer-to-peer traffic from a music download site is consuming a large amount of bandwidth. Which QoS mechanism can you implement to improve the network response time?

- A. Use CBWFQ to queue the peer-to-peer traffic into the default traffic class.
- B. Use class-based WRED to randomly drop the peer-to-peer traffic during network congestions.
- C. Use class-based policing to limit the peer-to-peer traffic rate.
- D. Use class-based shaping to delay any excessive peer-to-peer traffic.

Answer: C

Explanation:

QUESTION NO: 134

You are the network consultant from Cisco.com. Please point out two statements correctly describe an IPS device?

- A. It resembles a Layer 2 bridge.
- B. Traffic flow through the IPS resembles traffic flow through a Layer 3 router.
- C. Inline interfaces which have no IP addresses cannot be detected.
- D. Malicious packets that have been detected are allowed to pass through, but all subsequent traffic is blocked.

Answer: A,C

Explanation:

QUESTION NO: 135

Captain Marion's Videography delivers Internet digital video using 9 MPEG video encoders and a statistical multiplexer. Channels are packed into a 6-MHz channel bandwidth. The MPEG multiplexers monitor and allocate the appropriate bandwidth. The multiplexer measures available bandwidth and feeds back signaling to the MPEG encoders. Coding rates are then increased or decreased. Packet generation from each input source is controlled such that no packets are dropped and no extra null packets can be generated.

These bandwidth and traffic requirements work best with which mode of video delivery?

- A. fixed broadcast
- B. open looped
- C. quality equalization
- D. VoD delivery

Answer: A

Explanation:

QUESTION NO: 136

Please match the Cisco NAC appliance component to its description.

(1) Cisco NAS

(2)Cisco NAA

(3)Rule-set Lpdates

(4)Cisco NAM

(a) a centralized management point

(b) an in-band cr out-of-band device for network access control

(c) a Windows-based client which allows network access based on the tasks running

(d) a status crecker for operating systems,antivirus,antispysware,etc

A. (a)-(4);(b)-(1);(c)-(2);(d)-(3)

B. (a)-(3);(b)-(2);(c)-(4);(d)-(1)

C. (a)-(4);(b)-(3);(c)-(1);(d)-(2)

D. (a)-(2);(b)-(4);(c)-(3);(d)-(1)

Answer: A

Explanation:

QUESTION NO: 137

What is the first step that you would use Cisco Product Advisor for when selecting a router for an Edge solution?

A. determine types of protocols to be supported

B. determine the environment in which the router will be used

C. select the number of WAN ports required

D. select the number of LAN ports required

Answer: B

Explanation:

QUESTION NO: 138 DRAG DROP

Click and drag the Cisco client security software application on the left to its description on the right.

Cisco NAC Appliance Agent	client-side component provides device-based registry scans
Cisco Trust Agent	threat protector; application prevents malicious behavior before it can occur
Cisco Secure Services Client	single authentication framework works on multiple device types
Cisco Security Agent	application determines if security or management tools are installed and current

Answer:

Click and drag the Cisco client security software application on the left to its description on the right.

Cisco NAC Appliance Agent	Cisco NAC Appliance Agent
Cisco Trust Agent	Cisco Security Agent
Cisco Secure Services Client	Cisco Secure Services Client
Cisco Security Agent	Cisco Trust Agent

Explanation:

Click and drag the Cisco client security software application on the left to its description on the right.

Cisco NAC Appliance Agent	Cisco NAC Appliance Agent
Cisco Trust Agent	Cisco Security Agent
Cisco Secure Services Client	Cisco Secure Services Client
Cisco Security Agent	Cisco Trust Agent

QUESTION NO: 139

What is a criteria of the enterprise composite network model?

- A. includes all modules needed to meet any network design
- B. defines flexible boundaries between modules for scalability requirements
- C. clearly defines module boundaries and demarcation points to identify where traffic is
- D. requires specific core, distribution, and access layer requirements to match the model

Answer: C

Explanation:

QUESTION NO: 140 DRAG DROP

Click and drag the spanning-tree term on the left to its description on the right. You will not use all of the options

CST	identical to 802.1w
MST	assumes one spanning-tree instance for each bridged network regardless of number of VLANs
STP	provides a separate instance of 802.1w for each VLAN
RSTP	Cisco STP enhancement providing a separate 802.1D spanning tree instance for each VLAN
PVST+	original 802.1D version providing loop-free topology in a network with redundant links
rapid-PVST+	

Answer:

Click and drag the spanning-tree term on the left to its description on the right. You will not use all of the options

CST	RSTP
MST	CST
STP	rapid-PVST+
RSTP	PVST+
PVST+	STP
rapid-PVST+	

Explanation:

Click and drag the spanning-tree term on the left to its description on the right. You will not use all of the options

CST	RSTP
MST	CST
STP	rapid-PVSTP+
RSTP	PVSTP+
PVSTP+	STP
rapid-PVSTP+	

QUESTION NO: 141

Which routing protocol best fits these requirements?

- Supported by multiple router vendors
- Requires minimum router CPU and memory resources
- Uses a simple routing metric
- Supports manual or automatic route summarization

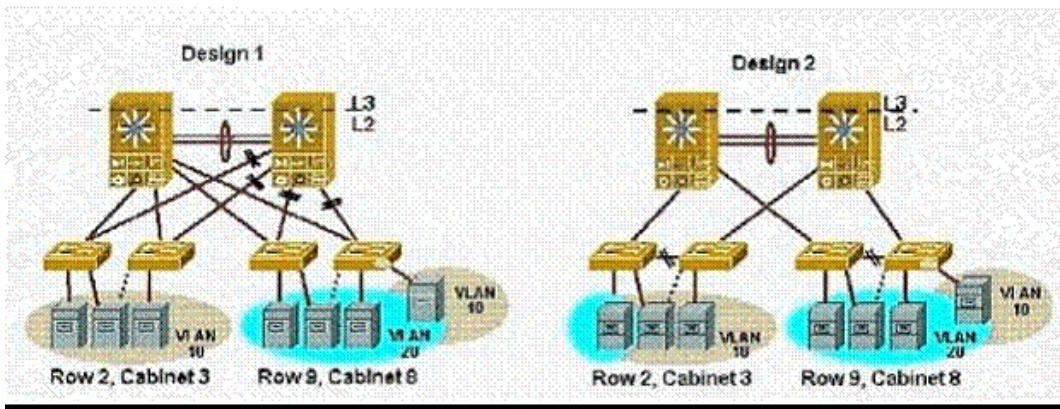
- A. EIGRP
- B. OSPF
- C. IS-IS
- D. RIPv2

Answer: D

Explanation:

QUESTION NO: 142

Refer to the exhibit. Which two statements about the topologies shown are correct? (Choose two.)



- A. Design 1 is a looped triangle design.
- B. Design 2 is a looped triangle design.
- C. Design 2 achieves quick convergence using RSTP.
- D. Both designs support stateful services at the aggregation layer.
- E. Design 2 is the most widely deployed in enterprise data centers.

Answer: A,D

Explanation:

QUESTION NO: 143

Which two of the following Cisco router platforms support Multicast Distributed Fast Switching? (Choose two.)

- A. 3600 series
- B. 7200 series with NSE-1
- C. 7500 series
- D. 12000 series

Answer: C,D

Explanation:

QUESTION NO: 144

Which two of these are characteristics of multicast routing? (Choose two.)

- A. multicast routing uses RPF.
- B. multicast routing is connectionless.
- C. In multicast routing, the source of a packet is known.
- D. When network topologies change, multicast distribution trees are not rebuilt, but use the original

path

E. Multicast routing is much like unicast routing, with the only difference being that it has a group of receivers rather than just one destination

Answer: A,C

Explanation:

QUESTION NO: 145

Which IOS QoS enhancement was created to address scalability and bandwidth guarantee issues?

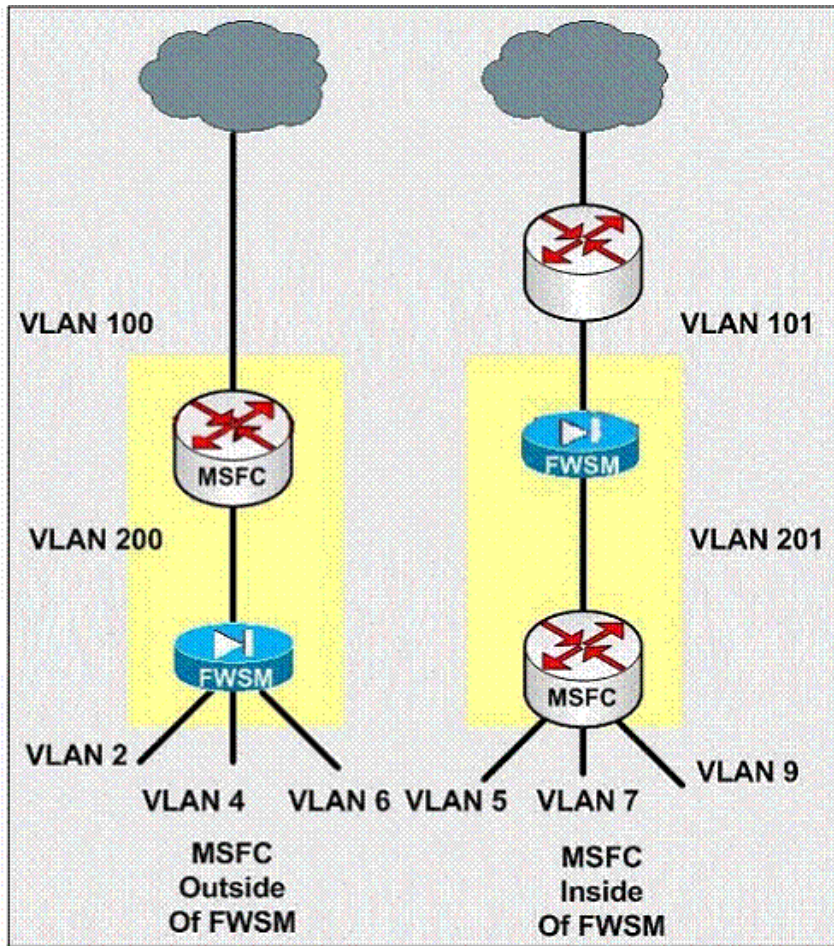
- A.** DiffServ
- B.** IntServ
- C.** RSVP
- D.** WFQ

Answer: C

Explanation:

QUESTION NO: 146

Refer to the exhibit. When deploying an MSFC and an FWSM, which statement is correct?



- A. Proper placement depends on the VLAN assignment.
- B. Place it outside the firewall.
- C. Place it inside the firewall to make design and management easier.
- D. Place it inside the firewall with multiple context modes connecting to all configured contexts.

Answer: A

Explanation:

QUESTION NO: 147

Sun Stable is a global insurance company with headquarters located in Houston, Texas. The campus there is made up of a number of office buildings located within the same vicinity. In 2003, a new building,

Building 331B was added. The additional building houses approximately 1000 employees. Rather than deploy a private branch exchange (PBX) in the new building, Sun Stable has decided to implement an IP telephony solution. External calls will be carried across a MAN link to another building, where a gateway connects into the worldwide PBX network of Sun Stable. Voice mail and

unified messaging components are required and all IP phones and workstations should be on separate VLANs and IP subnets.

Which IP telephony deployment best suits their need?

- A. single-site
- B. multisite with centralized call processing
- C. multisite with distributed call processing
- D. clustering over the WAN

Answer: A

Explanation:

QUESTION NO: 148

Which roaming option will keep them on the same IP subnet when client traffic is being bridged through LAN interfaces on two WLCs?

- A. Layer 1 intercontroller roaming
- B. Layer 2 intercontroller roaming
- C. Layer 3 intercontroller roaming
- D. Layer 4 intercontroller roaming

Answer: B

Explanation:

QUESTION NO: 149

Scalability is provided in the server farm module by which of the following design strategies?

- A. up to 10 Gbps of bandwidth at the access level
- B. redundant servers at the access level
- C. modular block design at the access level
- D. high port densities at the access level

Answer: C

Explanation:

QUESTION NO: 150

Which three of these are major scaling, sizing, and performance considerations for an IPsec design? (Choose three.)

- A. connection speed
- B. number of remote sites
- C. features to be supported
- D. types of devices at the remote site
- E. whether packets are encrypted using 3DES or AES
- F. number of routes in the routing table at the remote site

Answer: A,B,C

Explanation:

QUESTION NO: 151

Which three components comprise the AVVID framework? (Choose three.)

- A. common network infrastructure
- B. abstracted integration
- C. network solutions
- D. intelligent network services

Answer: A,C,D

Explanation:

QUESTION NO: 152

What is the term for a logical SAN which provides isolation among devices physically connected to the same fabric?

- A. ISL
- B. IVR
- C. VoQ
- D. VSANs
- E. Enhanced ISL

Answer: D

Explanation:

QUESTION NO: 153

Scalability is provided in the server farm module by which of the following design strategies?

- A. up to 10 Gbps of bandwidth at the access level
- B. redundant servers at the access level
- C. modular block design at the access level
- D. high port densities at the access level

Answer: C

Explanation:

QUESTION NO: 154

A company is using a multi-site centralized call processing model. Which feature ensures that the remote site IP phones will still have limited functionality given a WAN outage?

- A. Call Admission Control
- B. TAPI
- C. MGCP
- D. SRST

Answer: D

Explanation:

QUESTION NO: 155

As an experienced technician, you are responsible for Technical Support. A customer calls to ask the best signal level and noise level for cell phone. How to respond?

- A. -40dBm signal and -90dBm noise
- B. -50dBm signal and -90dBm noise
- C. -30dBm signal and -90dBm noise
- D. -20dBm signal and -90dBm noise

Answer: A

Explanation:

QUESTION NO: 156

As an experienced technician, you are responsible for Technical Support. Which of the following descriptions is correct about the characteristic of SLB one arm mode?

- A. This out-of-band approach supports scaling
- B. SLB is not inline.
- C. Mode is not as common as bridge or routed mode.
- D. Return traffic requires PBR, server default gateway pointing to SLB, or client source NAT.

Answer: C

Explanation:

QUESTION NO: 157

Which design topology incurs a performance penalty since there are two encryption-decryption cycles between any two remote sites?

- A. peer-to-peer
- B. peer-to-peer
- C. partial mesh
- D. hub and spoke
- E. full mesh

Answer: D

Explanation:

QUESTION NO: 158

Users at the Charleville Company began experiencing high network delays when Internet connectivity was enabled for all users. After investigating the traffic flow, you determine that peer-to-peer traffic from a music download site is consuming a large amount of bandwidth. Which QoS mechanism can you implement to improve the network response time?

- A. Use CBWFQ to queue the peer-to-peer traffic into the default traffic class.

- B. Use class-based WRED to randomly drop the peer-to-peer traffic during network congestions.
- C. Use class-based policing to limit the peer-to-peer traffic rate.
- D. Use class-based shaping to delay any excessive peer-to-peer traffic.

Answer: C

Explanation:

QUESTION NO: 159

Which statement about IDS/IPS design is correct?

- A. An IPS should be deployed if the security policy does not support the denial of traffic.
- B. An IPS analyzes a copy of the monitored traffic and not the actual forwarded packet.
- C. An IDS analyzes a copy of the monitored traffic and not the actual forwarded packet.
- D. Bandwidth considerations must be taken into account since IDS is deployed inline to traffic flow.

Answer: C

Explanation:

QUESTION NO: 160

What are disadvantages to storage directly attached to the application servers? (Choose three.)

- A. reliability
- B. scalability
- C. redundancy
- D. manageability

Answer: A,B,D

Explanation:

B: System administrators are faced with the challenging task to managing storage and making it scalable to accommodate future needs.

With storage directly attached to the server, scalability is difficult. The storage expansion capability is limited to the capacity of the server (for example, as measured by the number of I/O controllers and devices per controller configured is the server). The nature of the small computer system (SCSI) bus commonly used to connect commodity disks to a commodity server makes it difficult to allocate more disk storage without interrupting and rebooting the server, and thus affecting applications.

C: No redundancy is provided

Reference: Arch student guide p.13-6.

QUESTION NO: 161

As an experienced technician, you are responsible for infrastructure design and global configuration changes. You are asked to deploy a Voice over Wireless LAN for your company. If the cells have the same channel, the separation between them should be:

- A. 19dbm
- B. 10dbm
- C. 67dbm
- D. 86dbm

Answer: A

Explanation:

QUESTION NO: 162

When designing a converged network, which measures can be taken at the building access layer to help eliminate latency and ensure end-to-end quality of service can be maintained? (Choose three.)

- A. rate limit voice traffic
- B. configure spanning-tree for fast link convergence
- C. isolate voice traffic on separate VLANs
- D. classify and mark traffic close to the source

Answer: B,C,D

Explanation:

QUESTION NO: 163

Which two settings must be configured in order to use the GUI to configure Call Admission Control with voice applications? (Choose two.)

- A. QoS must be set to Platinum
- B. WMM must be enabled

- C. QoS must be set to Gold
- D. TSPEC must be disabled
- E. Cisco Compatible Extensions must be disabled

Answer: A,B

Explanation:

QUESTION NO: 164

Which IP telephony deployment model uses an H.225 Gatekeeper-Controlled trunk for call admission control within existing H.323 environments?

- A. single site with centralized call processing
- B. single site with distributed call processing
- C. multisite with centralized call processing
- D. multisite with distributed call processing

Answer: D

Explanation:

QUESTION NO: 165

You are the Cisco Network Designer in Cisco.com. In your company site, a NAS is both physically and logically in the traffic path. The NAS identifies clients solely based on their MAC addresses. In which access mode has this NAS been configured to operate?

- A. Layer 2 mode
- B. Layer 2 Edge mode
- C. Layer 3 mode
- D. Layer 3 In-Band mode

Answer: A

Explanation:

QUESTION NO: 166

The network administrator would like to generate synthetic traffic using the Service Assurance Agent contained in Cisco IOS. Which CiscoWorks network management application will be used to report the latency and availability for configured traffic operations on an end-to-end and hop-by-

hop (router-to-router) basis?

- A. nGenius Real-Time Monitor
- B. CiscoView
- C. Device Fault Manager
- D. Internetwork Performance Monitor

Answer: D

Explanation:

QUESTION NO: 167

A Fibre Channel fabric (or Fibre Channel switched fabric, FC-SW) is a switched fabric of Fibre Channel devices enabled by a Fibre Channel switch. Fabrics are normally subdivided by Fibre Channel zoning.

Each fabric has a name server and provides other services. Higher redundancy over FC-AL, P2P. Which path selection protocol is used by Fibre Channel fabrics?

- A. OSPF
- B. RIP
- C. FSPF
- D. VSANs

Answer: C

Explanation:

QUESTION NO: 168

Which two benefits does VoFR provide? (Choose two.)

- A. bandwidth efficiency
- B. cell-switching
- C. congestion notification
- D. heterogeneous network

Answer: A,C

Explanation:

QUESTION NO: 169

Which of these statements best describes VPLS?

- A. Neither broadcast nor multicast traffic is ever flooded in VPLS.
- B. Multicast traffic is flooded but broadcast traffic is not flooded in VPLS.
- C. VPLS emulates an Ethernet switch, with each EMS being analogous to a VLAN.
- D. Because U-PE devices act as IEEE 802.1 devices, the VPLS core must use STP.
- E. When the provider experiences an outage, IP re-routing restores PW connectivity and MAC re-learning is needed.

Answer: C

Explanation:

QUESTION NO: 170

When is the site-to-site remote access model appropriate? (Choose one.)

- A. for multiple ISDN connections
- B. for modem concentrated dial-up connections
- C. for a group of users in the same vicinity sharing a connection
- D. for use by mobile users

Answer: A

Explanation:

QUESTION NO: 171

VLAN Tagging, also known as Frame Tagging, is a method developed by Cisco to help identify packets travelling through trunk links. When an Ethernet frame traverses a trunk link, a special VLAN tag is added to the frame and sent across the trunk link .How does ERS use the VLAN tag?

- A. provide service internetworking
- B. support transparency for Layer 2 frames
- C. indicate destination as a connection identifier
- D. map to the DLCI in service internetworking

Answer: C

Explanation:

QUESTION NO: 172

What is one of the reasons that custom QoS ACLs are recommended over automatic QoS when configuring ports on a Catalyst 6500 for use with IP phones?

- A. 79xx IP phones do not automatically mark voice packets with non-zero DSCP values.
- B. 79xx IP phones do not mark protocol packets such as DHCP, DNS, or TFTP with non-zero DSCP values.
- C. 79xx IP phones do not mark voice packets with optimal DSCP values.
- D. 79xx IP phones use a custom protocol to communicate CDP information to the switch.

Answer: C

Explanation:

QUESTION NO: 173

Fibre Channel, or FC, is a gigabit-speed network technology primarily used for storage networking. Fibre Channel is standardized in the T11 Technical Committee of the InterNational Committee for Information Technology Standards (INCITS), an American National Standards Institute (ANSI) accredited standards committee.

Which two of these correctly describe Fibre Channel? (Choose two.)

- A. supports multiple protocols
- B. works only in a shared or loop environment
- C. allows addressing for up to 4 million nodes
- D. provides a high speed transport for SCSI payloads

Answer: A,D

Explanation:

QUESTION NO: 174

Which two are characteristics of RSVP? (Choose two.)

- A. RSVP itself provides bandwidth and delay guarantees.
- B. For RSVP to be end-to-end, all devices must support RSVP.

C. RSVP reservations are maintained by a centralized reservations server.

D. An RSVP compatible QoS mechanism must be used to implement guarantees according to RSVP reservations.

Answer: B,D

Explanation:

QUESTION NO: 175

The Cisco MDS 9000 Series Multilayer SAN Switches can help lower the total cost of ownership of the most demanding storage environments. By combining a robust and flexible hardware architecture with multiple layers of network and storage-management intelligence, the Cisco MDS 9000 Series helps you build highly available, scalable storage networks with advanced security and unified management.

What method does the Cisco MDS 9000 Series use to support trunking?

A. ISL

B. VLAN Trunk

C. VoQ

D. Enhanced ISL

Answer: D

Explanation:

QUESTION NO: 176

Which QoS requirement applies to streaming video traffic?

A. one-way latency of 150 ms to 200 ms

B. jitter of 30 ms or less

C. packet loss of 2 percent or less

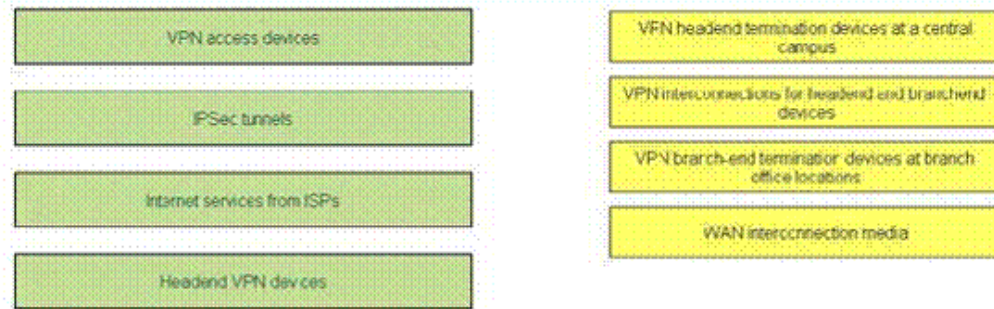
D. 150bps of overhead bandwidth

Answer: C

Explanation:

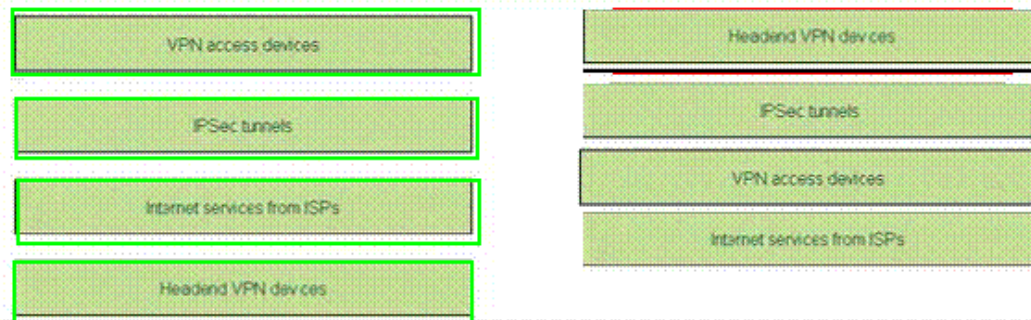
QUESTION NO: 177 DRAG DROP

Click and drag the VPN component on the left and drop it on its description on the right.

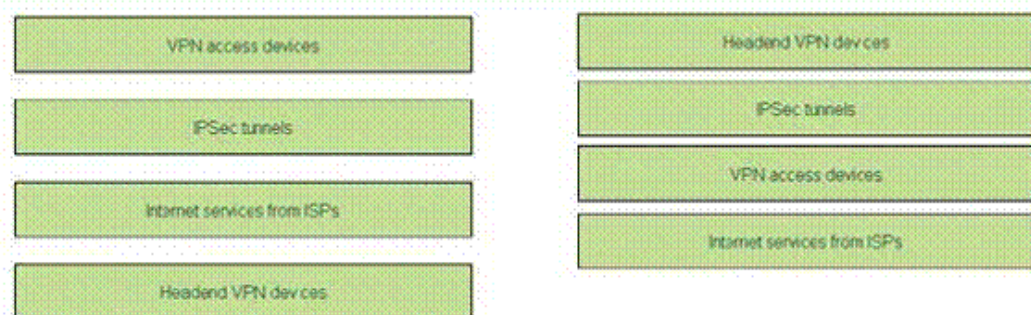


Answer:

Click and drag the VPN component on the left and drop it on its description on the right.



Explanation: Click and drag the VPN component on the left and drop it on its description on the right.



QUESTION NO: 178

To ensure voice packets are kept within the Committed Information Rate (CIR) of a Frame Relay link, what should be used in the CPE?

- A. prioritization
- B. classification
- C. fragmentation
- D. traffic shaping

Answer: D

Explanation:

QUESTION NO: 179

You are the Cisco Network Designer in Cisco.com. Which layer NAS operating mode are ACL filtering and bandwidth throttling only provided during posture assessment?

- A. Layer 2
- B. Layer 3
- C. Layer 4
- D. out-of-band

Answer: D

Explanation:

QUESTION NO: 180

A security analysis at The Potomac Canal Company recommends installing an IDS appliance and a firewall appliance. These appliances should connect directly into a Layer 3 switch. A load balancer and SSL termination have also been recommended. Potomac's management have expressed concern over the cost.

You suggest using integrated blades. What is one advantage and one disadvantage of your design proposal? (Choose two.)

- A. The data center would need several devices to achieve its goal.
- B. Increased usage of standalone devices is cost-effective.
- C. Using integrated blades would only require two devices.
- D. Putting all security devices in a single chassis provides a single point of failure.

Answer: C,D

Explanation:

Topic 4, Volume D

QUESTION NO: 181

Which technology allows centralized storage services to be shared across different VSANs?

- A. IVR
- B. FSPF
- C. FICON
- D. SANTap

Answer: A

Explanation:

QUESTION NO: 182

Which content networking device allows bandwidth configuration settings so that streaming content will not interfere with other network traffic?

- A. IP/TV Control Server
- B. Content Distribution Manager
- C. Content Engine
- D. IP/TV Broadcast Server

Answer: A

Explanation:

QUESTION NO: 183

What is the purpose of IGMP in a multicast implementation?

- A. it is not used in multicast
- B. it determines the virtual address group for a multicast destination
- C. it dynamically registers individual hosts in a multicast group on a specific LAN
- D. it is used on WAN connections to determine the maximum bandwidth of a connection
- E. it determines whether Bidirectional PIM or PIM sparse mode will be used for a multicast flow

Answer: C

Explanation:

QUESTION NO: 184

What is high availability?

- A. redundant infrastructure
- B. clustering of computer systems
- C. reduced MTBF
- D. continuous operation of computing systems

Answer: D

Explanation:

QUESTION NO: 185

Which two characteristics are most typical of a SAN? (Choose two.)

- A. NICs are used for network connectivity.
- B. Servers request specific blocks of data.
- C. Storage devices are directly connected to servers.
- D. A fabric is used as the hardware for connecting servers to storage devices.
- E. The TCO is higher because of the cost of director class storage switches.

Answer: B,D

Explanation:

QUESTION NO: 186

Which VPN management feature would be considered to ensure that the network had the least disruption of service when making topology changes?

- A. dynamic reconfiguration
- B. path MTU discovery
- C. auto setup
- D. remote management

Answer: A

Explanation:

Dynamic reconfiguration: All configuration changes should take effect without requiring a reboot of the device. Disruption of service with a fully loaded VPN device can potentially impact thousands of individual users.

Reference: Arch student guide p.9-17

QUESTION NO: 187

You are the Cisco Network Designer in Cisco.com. Which of these are important when determining how many users a NAS can support?

- A. bandwidth
- B. number of plug-ins per scan
- C. total number of network devices
- D. number of checks in each posture assessment

Answer: B,C,D

Explanation:

QUESTION NO: 188

Acme Costume Company is connecting its manufacturing facilities to its stores with a small point-to-multipoint Frame Relay IP WAN. Little growth is expected in the network infrastructure. Up to this point the company has been using a dial-on-demand network. Dropping WAN costs, however, have led them to consider using a high-speed WAN solution to improve access. Which two routing protocols could you deploy to support the new larger network while keeping costs down? (Choose two.)

- A. RIP
- B. RIPv2
- C. EIGRP
- D. OSPF

Answer: C,D

Explanation:

QUESTION NO: 189

The VPN termination function provides the ability to connect two networks together securely over the internet. Which of these is true of IP addressing with regard to VPN termination?

- A. termination devices need routable addresses inside the VPN

- B. termination devices need not routable addresses inside the VPN
- C. IGP routing protocols will update their routing tables over an IPsec VPN
- D. addressing designs need to allow for summarization

Answer: D

Explanation:

QUESTION NO: 190

When dealing with transparent caching, where should the Content Engines be placed?

- A. close to the servers
- B. close to the end users
- C. at the Internet edge
- D. in front of web server farms

Answer: B

Explanation:

QUESTION NO: 191

Which of these statements is true of routing protocols in a hub-and-spoke IPsec VPN topology?

- A. EIGRP can summarize per interface.
- B. OSPF router databases remain independent.
- C. When they are configured with stubs, EIGRP regularly floods the topology.
- D. OSPF topology decisions are made independent of hierarchy or area.

Answer: A

Explanation:

QUESTION NO: 192

Which three things can be restricted by the Class of Service in a traditional PBX? (Choose three.)

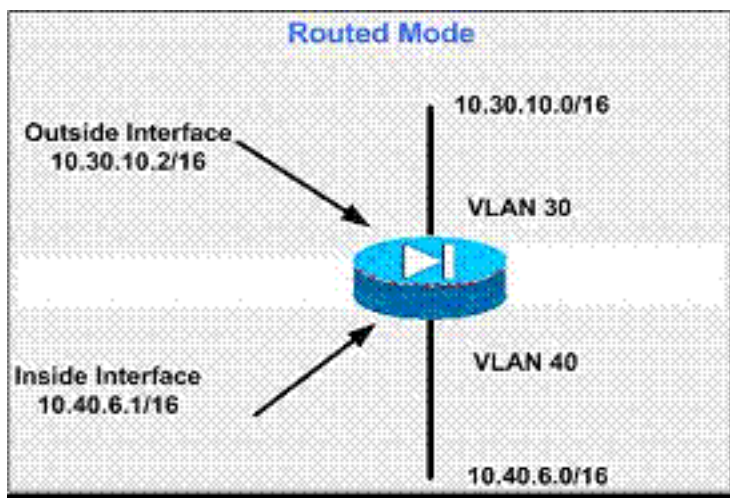
- A. dial plans
- B. dialed numbers
- C. voice mail prompts
- D. phone features

Answer: A,B,D

Explanation:

QUESTION NO: 193

Which two characteristics are true of a firewall running in routed mode based on the following information?



- A. FWSM routes traffic between the VLANs.
- B. FWSM switches traffic between the VLANs.
- C. Routed mode is often called bump-in-the-wire mode.
- D. Routed mode firewall deployments are used most often in current designs.

Answer: A,D

Explanation:

QUESTION NO: 194

Which statement about CiscoWorks 2000 Inventory Manager is true?

- A. It uses SNMP v1.
- B. It scans devices for hardware information.

- C. It scans and records the operational status of devices.
- D. When the configuration of a device changes, the inventory is automatically updated.

Answer: B

Explanation:

QUESTION NO: 195

You are the Cisco Network Designer in Cisco.com. Which of these practices should you follow when designing a Layer 3 routing protocol?

- A. Never peer on transit links.
- B. Build squares for deterministic convergence.
- C. Build inverted U designs for deterministic convergence.
- D. Summarize routes at the distribution to the core to limit EIGRP queries or OSPF LSA propagation.

Answer: D

Explanation:

QUESTION NO: 196

Which two statements are true about MLP interleaving? (Choose two.)

- A. It fragments and encapsulates all packets in a fragmentation header.
- B. Packets smaller than the fragmentation size are interleaved between the fragments of the larger packets.
- C. Packets larger than the fragmentation size are always fragmented, and cannot be interleaved, even if the traffic is voice traffic.
- D. It fragments and encapsulates packets that are longer than a configured size, but does not encapsulate smaller packets inside a fragmentation header.

Answer: B,D

Explanation:

Previous implementations of Cisco IOS Multilink PPP (MLP) include support for Link Fragmentation Interleaving (LFI). This feature allows the delivery of delay-sensitive packets, such as the packets of a Voice call, to be expedited by omitting the PPP Multilink Protocol header and sending the packets as raw PPP packets in between the fragments of larger data packets. This feature works well on bundles consisting of a single link. However, when the bundle contains

multiple links there is no way to keep the interleaved packets in sequence with respect to each other.

Interleaving on MLP allows large packets to be multilink encapsulated and fragmented into a small enough size to satisfy the delay requirements of real-time traffic; small real-time packets are not multilink encapsulated and are transmitted between fragments of the large packets.

Note: The following URL from Cisco's website explains this feature:

http://www.cisco.com/univercd/cc/td/doc/product/software/ios122/122cgcr/fqos_c/fqcprt6/qcflfi.htm#wp1000907

"(Optional) Configures a maximum fragment delay. If, for example, you want a voice stream to have a maximum bound on delay of 20 milliseconds (ms) and you specify 20 ms using this command, MLP will choose a fragment size based on the configured value."
Packets are fragmented when they exceed the configured maximum delay.

QUESTION NO: 197

_____ dBm is the recommended radius of a cell for a voice-ready wireless network.

- A. 4
- B. 6
- C. 7
- D. 10
- E. 67

Answer: E

Explanation: In "Designing Cisco Network Service Architectures (ARCH)", page 509

The radius of the cell should be **-67 dBm**.

QUESTION NO: 198

What are two considerations to using IP Multicast delivery? (Choose two.)

- A. no congestion avoidance
- B. not for bandwidth intensive applications
- C. no guaranteed delivery mechanism
- D. source sends multiple data streams out each interface

Answer: A,C

Explanation: Explanation; Multicast disadvantage are Best-effort delivery, No congestion avoidance, Duplicates and Out-of order delivery.

QUESTION NO: 199

The Cisco IOS SLB feature is a Cisco IOS-based solution that provides server load balancing. This feature allows you to define a virtual server that represents a cluster of real servers, known as a server farm.

When a client initiates a connection to the virtual server, the IOS SLB load balances the connection to a chosen real server, depending on the configured load balance algorithm or predictor.

Which three implementation modes may be used to deploy SLB? (Choose three.)

- A. Router mode
- B. One-arm mode
- C. Three-arm mode
- D. Bridge mode inline

Answer: A,B,D

Explanation:

QUESTION NO: 200

With Call Manager v3.1, what is the maximum number of servers in a Cluster?

- A. 3
- B. 6
- C. 7
- D. 8

Answer: D

Explanation:

The primary advantage of the distributed call processing model is that, by using local call processing, it provides the same level of features and capabilities whether the IP WAN is available or not. Each site can have from one to eight Cisco CallManager servers in a cluster based on the number of users.

Reference: Arch student guide p.11-43

QUESTION NO: 201

You are the Cisco Network Designer in Cisco.com. You are designing an e-Commerce module, which routing statement is correct?

- A.** Routing is mostly static.
- B.** Hardcoded IP addresses are used to support failover.
- C.** Inbound servers use the CSM or ACE as the default gateway.
- D.** VLANs between the access layer switches are used for FHRP protocols.

Answer: A

Explanation:

QUESTION NO: 202

A network vulnerability scanner is part of which critical element of network and system security?

- A.** host security
- B.** perimeter security
- C.** security monitoring
- D.** policy management

Answer: C

Explanation:

QUESTION NO: 203

You are the Cisco Network Designer in Cisco.com. What is the term for a logical SAN which provides isolation among devices physically connected to the same fabric?

- A. InterSwitch Link
- B. Virtua LAN
- C. Virtual Output Queuing
- D. virtual storage area network

Answer: D

Explanation:

QUESTION NO: 204

Scalability is provided in the server farm module by which of the following design strategies?

- A. up to 10 Gbps of bandwidth at the access level
- B. redundant servers at the access level
- C. modular block design at the access level
- D. high port densities at the access level

Answer: C

Explanation:

QUESTION NO: 205

A virtual storage area network (VSAN) is a collection of ports from a set of connected Fibre Channel switches, that form a virtual fabric. Which technology allows centralized storage services to be shared across different VSANs?

- A. IVR
- B. FSPF
- C. FICON
- D. SANTap

Answer: A

Explanation:

QUESTION NO: 206

What four functions does Web Cache Communication Protocol (WCCP) incorporate? (Choose

four.)

- A. load balancing
- B. scalability
- C. remote management
- D. fault tolerance
- E. service assurance

Answer: A,B,D,E

Explanation:

QUESTION NO: 207

Which of the following is the primary consideration to scale VPNs?

- A. packets per second
- B. number of remote sites
- C. throughput bandwidth
- D. number of tunnels

Answer: A

Explanation:

QUESTION NO: 208

Lafayette Productions is looking for a new ISP that has improved availability, load balancing, and catastrophe protection. Which type of ISP connectivity solution would be best?

- A. single run
- B. multi-homed
- C. stub domain EBGP
- D. direct BGP peering

Answer: B

Explanation:

QUESTION NO: 209

In a base e-Commerce module design, which routing statement is correct?

- A. Routing is mostly static.
- B. Hardcoded IP addresses are used to support failover.
- C. Inbound servers use the CSM or ACE as the default gateway.
- D. VLANs between the access layer switches are used for FHRP protocols.

Answer: A

Explanation:

QUESTION NO: 210

In which tunnel-less VPN topology do group members register with a key server in order to receive the security association necessary to communicate with the group?

- A. Easy VPN
- B. GRE tunneling
- C. Virtual Tunnel Interfaces
- D. Dynamic Multipoint VPN
- E. Group Encrypted Transport VPN

Answer: E

Explanation:

QUESTION NO: 211

Which two of these are advantages of placing the VPN device in the DMZ on the firewall? (Choose two.)

- A. fewer devices to manage
- B. moderate-to-high scalability
- C. stateful inspection of decrypted VPN traffic
- D. increased bandwidth with additional interfaces
- E. decreased complexity as traffic is filtered from the firewall

Answer: B,C

Explanation:

QUESTION NO: 212

Under which two circumstances should Spanning Tree Protocol be implemented? (Choose two.)

- A. to ensure a loop-free topology
- B. to protect against user-side loops
- C. when a VLAN spans access layer switches
- D. for the most deterministic and highly available network topology
- E. because of the risk of lost connectivity without Spanning Tree Protocol

Answer: B,C

Explanation:

QUESTION NO: 213

Which two of these are advantages of placing the VPN device parallel to the firewall? (Choose two.)

- A. high scalability
- B. the design supports a layered security model
- C. firewall addressing does not need to change
- D. IPsec decrypted traffic is inspected by the firewall
- E. there is a centralized point for logging and content inspection

Answer: A,C

Explanation:

QUESTION NO: 214

What will an Easy VPN hardware client require in order to insert its protected network address when it connects using network extension mode?

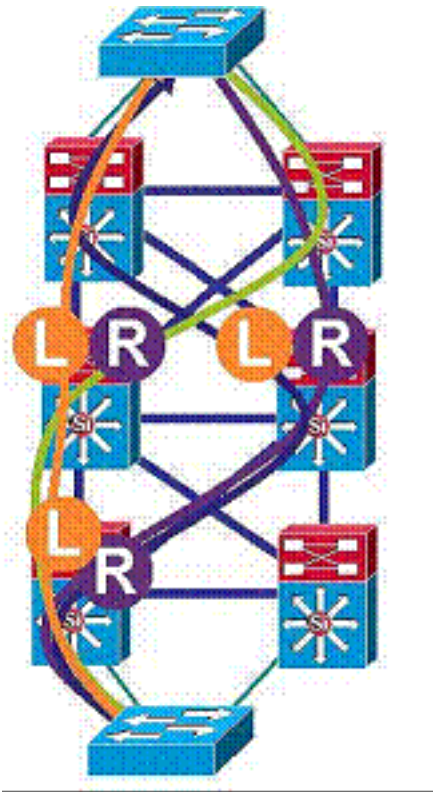
- A. RADIUS or LDAP
- B. an internal router running EIGRP
- C. Reverse Route Injection and OSPF or RIPv2
- D. the VPN appliance to be deployed in line with the firewall

Answer: C

Explanation:

QUESTION NO: 215

Which two practices will avoid Cisco Express Forwarding polarization?(Choose two.)



- A. The core layer should use default Layer 3 hash information.
- B. The core layer should use default Layer 4 hash information.
- C. The distribution layer should use default Layer 3 hash information.
- D. The distribution layer should use default Layer 4 hash information.
- E. The core layer should use Layer 3 and Layer 4 information as input to the Cisco Expressing Forwarding hashing algorithm.
- F. The distribution layer should use Layer 3 and Layer 4 information as input into the Cisco Expressing Forwarding hashing algorithm.

Answer: A,F

Explanation:

QUESTION NO: 216

When a router has to make a rate transition from LAN to WAN, what type of congestion needs should be considered in the network design?

- A. RX-queue deferred

- B. TX-queue deferred
- C. RX-queue saturation
- D. TX-queue saturation
- E. RX-queue starvation
- F. TX-queue starvation

Answer: F

Explanation:

QUESTION NO: 217

What is the recommended practice when considering VPN termination and firewall placement?

- A. have the firewall and VPN appliance deployed in parallel
- B. place the VPN in line with the firewall, with the VPN terminating inside the firewall
- C. place the public side of the VPN termination device in the DMZ behind a firewall
- D. place the VPN in line with the firewall, with the VPN terminating outside the firewall

Answer: C

Explanation:

QUESTION NO: 218

Which of these statements is correct regarding SSO and Cisco NSF?

- A. Utilizing Cisco NSF in Layer 2 environments can reduce outages to one to three seconds.
- B. Utilizing SSO in Layer 3 environments can reduce outages to one to three seconds.
- C. Distribution switches are single points of failure causing outages for the end devices.
- D. Utilizing Cisco NSF and SSO in a Layer 2 environment can reduce outages to less than one second.
- E. NSF and SSO with redundant supervisors have the most impact on outages at the access layer.

Answer: E

Explanation:

QUESTION NO: 219

Which of these is a correct description of SSO?

- A. It will only become active after a software failure.
- B. It will only become active after a hardware failure.
- C. It requires that Cisco NSF be enabled in order to work successfully.
- D. It synchronizes the MAC, FIB, and adjacency tables between Active and Standby Route Processors.

Answer: D

Explanation:

QUESTION NO: 220

Which of these recommended designs provides the highest availability?

- A. map the Layer 2 VLAN number to the Layer 3 subnet
- B. control route propagation to edge switches using distribute lists
- C. use a Layer 2 distribution interconnection link with HSRP or GLBP
- D. use a Layer 3 distribution interconnection link with HSRP or GLBP
- E. use equal-cost Layer 3 load balancing on all links to limit the scope of queries in EIGRP

Answer: A

Explanation:

QUESTION NO: 221

An organization hires a contractor who only needs access to email and a group calendar. They do not need administrator access to the computer. Which VPN model is the most appropriate?

- A. Thin Model
- B. Thick Client
- C. Port Forwarding
- D. Clientless Access
- E. Layer 3 Network Access

Answer: D

Explanation:

QUESTION NO: 222

In which NAS operating mode are ACL filtering and bandwidth throttling only provided during posture assessment?

- A. Layer 2
- B. Layer 3
- C. in-band
- D. out-of-band
- E. edge
- F. central

Answer: D

Explanation:

QUESTION NO: 223

Which two of these are recommended practices with trunks? (Choose two.)

- A. use ISL encapsulation
- B. use 802.1q encapsulation
- C. set ISL to desirable and auto with encapsulation negotiate to support ILS protocol negotiation
- D. use VTP server mode to support dynamic propagation of VLAN information across the network
- E. set DTP to desirable and desirable with encapsulation negotiate to support DTP protocol negotiation.

Answer: B,E

Explanation:

QUESTION NO: 224

Which of these is a benefit of using Network Admission Control instead of Cisco Identity Based Networking Services?

- A. NAC can authenticate using 802.1X and IBNS cannot
- B. NAC can ensure only compliant machines connect and IBNS cannot
- C. NAC can ensure access to the correct network resources and IBNS cannot
- D. NAC can manage user mobility and reduce overhead costs and IBNS cannot

Answer: B

Explanation:

QUESTION NO: 225

Which three of these Metro Ethernet services map to E-Line services that are defined by the MEF? (Choose three.)

- A. Ethernet Private Line
- B. Ethernet Wire Service
- C. Ethernet Relay Service
- D. Ethernet Multipoint Service
- E. Ethernet Relay Multipoint Service

Answer: A,B,C

Explanation:

QUESTION NO: 226

Which two of these Metro Internet services map to E-LAN services that are defined by the MEF? (Choose two.)

- A. Ethernet Private Line
- B. Ethernet Wire Service
- C. Ethernet Relay Service
- D. Ethernet Multipoint Service
- E. Ethernet Relay Multipoint Service

Answer: D,E

Explanation:

QUESTION NO: 227

Which two of these are characteristics of Metro Ethernet? (Choose two.)

- A. class of service
- B. bandwidth profiles
- C. user-network interface
- D. Ethernet LAN circuit attributes
- E. Ethernet virtual circuit attributes

Answer: C,E

Explanation:

QUESTION NO: 228

Which three of these are important when determining NAS Server scaling? (Choose three.)

- A. interface bandwidth
- B. rescan timer interval
- C. total number of network devices
- D. number of new user authentications per second
- E. which operating system is loaded on the client
- F. number of checks performed in a posture assessment

Answer: B,D,F

Explanation:

QUESTION NO: 229

Which of these is true of a Layer 3 out-of-band NAS deployment?

- A. The NAS acts as a gateway for all Layer 3 traffic.
- B. Only the MAC address is used to identify the client device.
- C. User traffic remains on the same VLAN for the duration of the connection.
- D. After authentication and posture assessment, client traffic no longer passes through the NAS.

Answer: D

Explanation:

Topic 5, Volume E

QUESTION NO: 230

Your MPLS implementation is currently using internal backdoor links. What can you do to minimize the impact of having these links?

- A. use BGP as the CE-PE routing protocol
- B. use OSPF as the CE-PE routing protocol
- C. use EIGRP as the CE-PE routing protocol
- D. use the SP to redistribute routes as external routes for OSPF and EIGRP
- E. use route redistribution at each location to ensure external routes are imported into the IGP

Answer: A

Explanation:

QUESTION NO: 231

One of your customers wishes to use the NAS to perform DHCP functions and does not currently have a Layer 3 gateway in its production network. Which gateway mode is appropriate for this customer?

- A. Virtual Gateway
- B. Real-IP Gateway
- C. NAT Gateway
- D. IP-IP Gateway

Answer: B

Explanation:

QUESTION NO: 232

Which of these is a Layer 2 transport architecture that provides packet-based transmission optimized for data based on a dual (counter-rotating) ring topology?

- A. DTP
- B. RPR
- C. SDH
- D. CWDM
- E. DWDM

Answer: B

Explanation:

QUESTION NO: 233

Which of these is a benefit of ESM?

- A. supports multiple MIBs
- B. includes NetFlow, NBAR, and IP SLA software subsystems
- C. includes NetFlow, syslog, and IP SLA software subsystems
- D. includes a predefined framework for filtering and correlating messages
- E. supports two logging processes so output can be sent in standard and ESM format

Answer: D

Explanation:

QUESTION NO: 234

Which of these ports does syslog use to send messages to a syslog server?

- A. TCP 502
- B. TCP 514
- C. TCP 520
- D. UDP 502
- E. UDP 514
- F. UDP 520

Answer: E

Explanation:

QUESTION NO: 235

To which of these does IP multicast send packets?

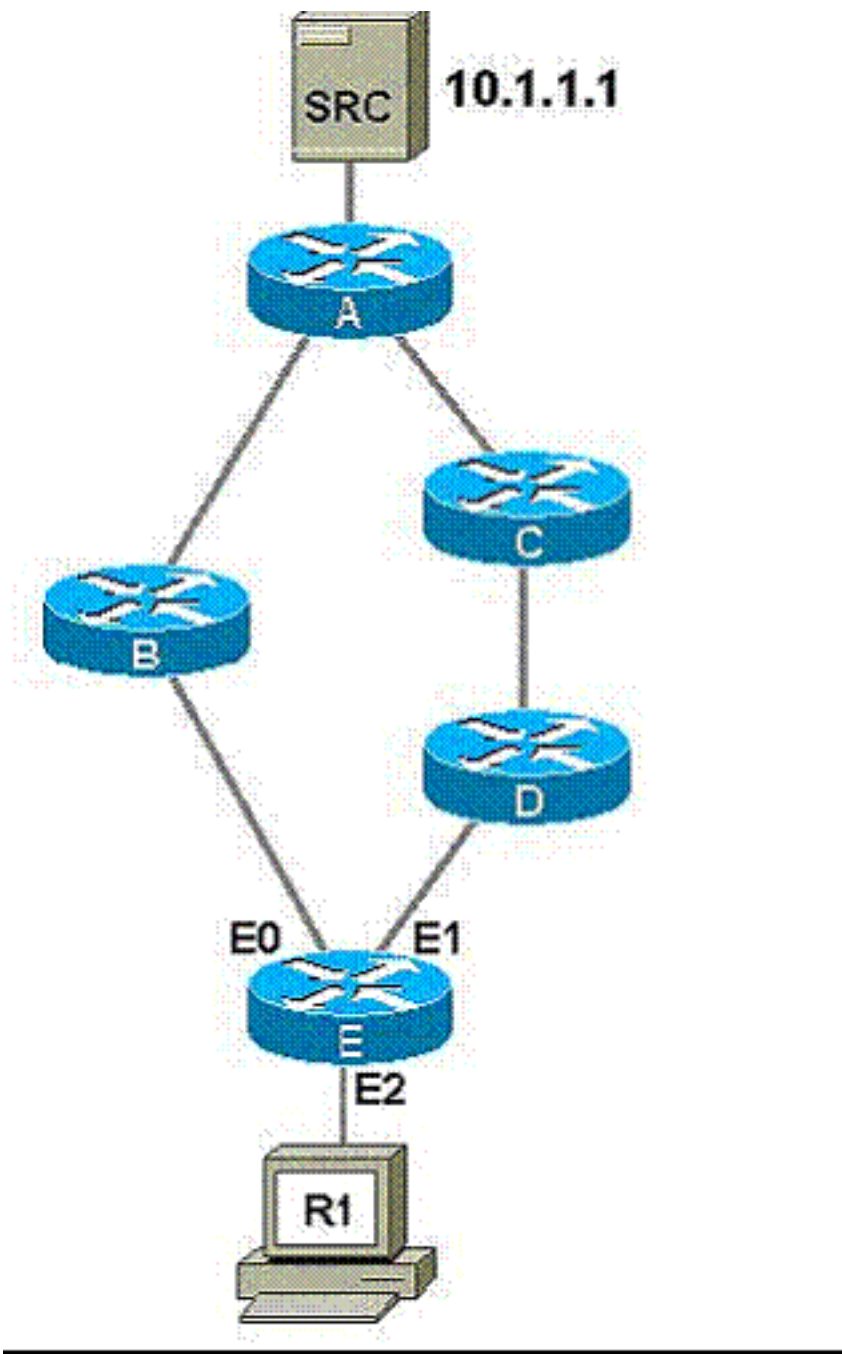
- A. a single host
- B. a subset of hosts
- C. all hosts sequentially
- D. all hosts simultaneously

Answer: B

Explanation:

QUESTION NO: 236

Refer to the exhibit. Which two statements are correct regarding the creation of a multicast distribution tree? (Choose two.)



- A. Each router determines where to send the JOIN request.
- B. The tree will be built based on the IP address of the E2 interface on router E.
- C. The best path to the source will be discovered in the unicast routing table on router B.

- D. The best path to the source will be discovered in the unicast routing table on router C.
- E. The best path to the source will be discovered in the unicast routing table on router E.

Answer: A,E

Explanation:

QUESTION NO: 237

What is the default value of the SPT threshold in Cisco routers?

- A. 0
- B. 1
- C. 2
- D. 4
- E. 16
- F. infinity

Answer: A

Explanation:

QUESTION NO: 238

Which two of these multicast deployments are most susceptible to attacks from unknown sources?
(Choose two.)

- A. ASM
- B. BiDir PIM
- C. PIM-SM RP
- D. RP-Switchover
- E. Source Specific Multicast

Answer: A,B

Explanation:

QUESTION NO: 239

Which of the following is a characteristic of a data center core?

- A. Server-to-server traffic always remains in the core layer.

- B. The recommended practice is for the core infrastructure to be in Layer 3.
- C. The boundary between Layer 2 and Layer 3 should be implemented in the aggregation layer.
- D. The Cisco Express Forwarding hashing algorithm is the default, based on the IP address and Layer 4 port.
- E. Core layer should run BGP along with an IGP because iBGP has a lower administrative distance than any IGP.

Answer: B

Explanation:

QUESTION NO: 240

Which two design recommendations are most appropriate when OSPF is the data center core routing protocol? (Choose two.)

- A. Never use passive interfaces.
- B. Use NSSA areas from the core down.
- C. Use totally stub areas to stop type 3 LSAs.
- D. Use the lowest Ethernet interface IP address as the router ID.
- E. Tune OSPF timers to enable OSPF to achieve quicker convergence

Answer: B,E

Explanation:

QUESTION NO: 241

Which two design recommendations are most appropriate when EIGRP is the data center core routing protocol? (Choose two.)

- A. Summarize data center subnets.
- B. Use passive interfaces to ensure appropriate adjacencies.
- C. Tune the EIGRP timers to enable EIGRP to achieve quicker convergence.
- D. Adjust the default bandwidth value to ensure proper bandwidth on all links.
- E. Advertise a default summary route into the data center core from the aggregation layer.

Answer: A,E

Explanation:

QUESTION NO: 242

Which two statements correctly describe a situation in which an Active/Standby Service Module

design is being used? (Choose two.)

- A. Troubleshooting is more complicated.
- B. Service and switch modules are underutilized.
- C. Layer 2 adjacency is required with the servers that use this design.
- D. Layer 3 adjacency is required with the servers that use this design.
- E. Load balancing will always occur across both access layer uplinks.

Answer: B,C

Explanation:

QUESTION NO: 243

Which statement correctly describes a situation in which VRFs are used in the data center?

- A. Partitioning of network resources is enabled.
- B. VRFs cannot support path isolation from MAN/WAN designs.
- C. VRFs cannot be used to map a virtualized data center to a MPLS implementation.
- D. VRFs do not allow for the use of application services with multiple access topologies.
- E. An access design using a VRF allows for an aggregation layer service module solution.

Answer: A

Explanation:

QUESTION NO: 244

Which statement about data center access layer design modes is correct?

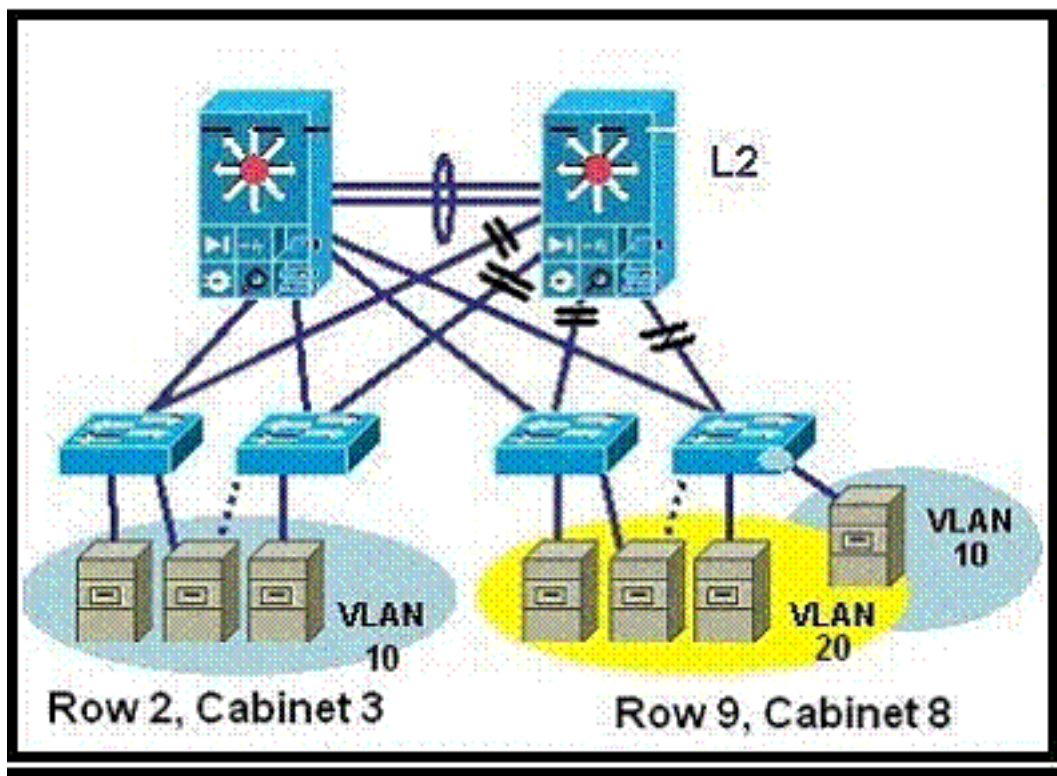
- A. The access layer is the first oversubscription point in a data center design.
- B. When using a Layer 2 loop-free design, VLANs are extended into the aggregation layer.
- C. When using a Layer 2 looped design, VLANs are not extended into the aggregation layer.
- D. When using a Layer 3 design, stateful services requiring Layer 2 connectivity are provisioned from the aggregation layer.
- E. The data center access layer provides the physical-level connections to the server resources and only operates at Layer 3.

Answer: A

Explanation:

QUESTION NO: 245

Refer to the exhibit. Which statement is correct regarding the topology shown?



- A. It achieves quick convergence with 802.1w/s.
- B. It is currently the most widely deployed in enterprise data centers.
- C. It is a looped square that achieves resiliency with dual homing and STP.
- D. It is a looped triangle that achieves resiliency with dual homing and STP.

Answer: B

Explanation:

QUESTION NO: 246

Which two statements about Network Attached Storage are correct? (Choose two.)

- A. Data is accessed using NFS or CIFS.
- B. Data is accessed at the block level.
- C. NAS is referred to as captive storage.
- D. Storage devices can be shared between users.
- E. A NAS implementation is not as fast as a DAS implementation.

Answer: B,E

Explanation:

QUESTION NO: 247

Which two of these correctly describe Fibre Channel? (Choose two.)

- A. supports multiple protocols
- B. works only in a shared or loop environment
- C. allows addressing for up to 4 million nodes
- D. allows addressing for up to 8 million nodes
- E. provides a high speed transport for SCSI payloads
- F. may stretch to a distance of up to 100 km before needing extenders

Answer: A,E

Explanation:

QUESTION NO: 248

Which statement about Fibre Channel communications is correct?

- A. It operates much like TCP.
- B. Flow control is only provided by QoS.
- C. It must be implemented in an arbitrated loop.
- D. Communication methods are similar to those of an Ethernet bus.
- E. N_Port to N_Port connections use logical node connection points.

Answer: E

Explanation:

QUESTION NO: 249

What is the term for a logical SAN which provides isolation among devices physically connected to the same fabric?

- A. ISL
- B. IVR
- C. VoQ
- D. VSANs
- E. Enhanced ISL

Answer: D

Explanation:

QUESTION NO: 250

Which path selection protocol is used by Fibre Channel fabrics?

- A. IVR
- B. VoQ
- C. FSPF
- D. VSANs
- E. SANTap

Answer: C

Explanation:

QUESTION NO: 251

In a collapsed core design, which two benefits are provided by a second-generation Cisco MDS director? (Choose two.)

- A. a higher fan-out ratio
- B. fully redundant switches
- C. 100 percent port efficiency
- D. all ISLs contained within a single chassis
- E. higher latency and throughput than a core-edge design switch

Answer: B,C

Explanation:

QUESTION NO: 252

Which two statements about FCIP and iSCSI are correct? (Choose two.)

- A. The FCIP stack supports file-level storage for remote devices.
- B. Both require high throughput with low latency and low jitter.
- C. The purpose of FCIP is to provide connectivity between host and storage.
- D. The iSCSI stack supports block-level storage for remote devices.
- E. The purpose of iSCSI is to provide connectivity between separate wide-area SANs.

Answer: B,D

Explanation:

QUESTION NO: 253

One of your customers has deployed a Layer 3 gateway in the untrusted network. Which gateway mode is appropriate for this customer?

- A. Virtual Gateway
- B. Real-IP Gateway
- C. NAT Gateway
- D. Central Gateway

Answer: A

Explanation:

QUESTION NO: 254

Which two statements about zoning are correct? (Choose two.)

- A. Zoning increases security.
- B. DNS queries are used for software zoning.
- C. Software zoning is more secure than hardware zoning.
- D. When using zones and VSANs together, the zone is created first.
- E. Zoning requires that VSANs be established before it becomes operational.

Answer: A,B

Explanation:

QUESTION NO: 255

At a certain customer's site, a NAS is logically in the traffic path but not physically in the traffic path. The NAS identifies clients by their IP addresses. In which access mode has this NAS been configured to operate?

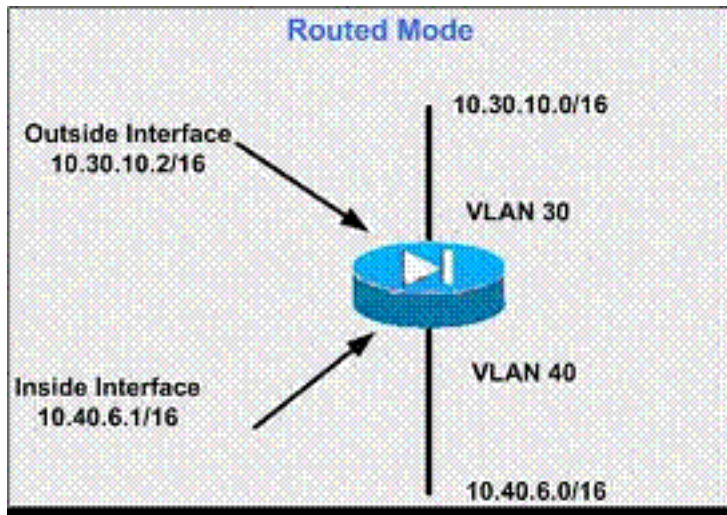
- A. Layer 2 Edge mode
- B. Layer 2 Central mode
- C. Layer 2 In-Band mode
- D. Layer 3 mode

Answer: D

Explanation:

QUESTION NO: 256

Refer to the exhibit. Which two of these are characteristics of a firewall running in transparent mode? (Choose two.)



- A. FWSM routes traffic between the VLANs.
- B. FWSM switches traffic between the VLANs.
- C. Transparent mode is often called bump-in-the-wire mode.
- D. Transparent mode firewall deployments are used most often in current designs.
- E. Traffic routed between VLANs is subject to state tracking and other firewall configurable options.

Answer: B,C

Explanation:

QUESTION NO: 257

At a certain customer's site, a NAS is both physically and logically in the traffic path. The NAS identifies clients solely based on their MAC addresses. In which access mode has this NAS been configured to operate?

- A. Layer 2 mode
- B. Layer 3 Edge mode
- C. Layer 3 Central mode
- D. Layer 3 In-Band mode

Answer: A

Explanation:

QUESTION NO: 258

What are two characteristics of the SLB One-arm mode? (Choose two.)

- A. It is not as common as bridge mode.
- B. The MSFC is not directly connected to the CSM.
- C. Outbound traffic from servers may need to be directed by PBR or CSNAT to the CSM.
- D. The SLB is moved to a position where selected inbound and outbound server traffic goes through the SLB.
- E. The CSM statically routes inbound server traffic to the aggregation switch FWSM, then to the connected server subnet.

Answer: A,C

Explanation:

QUESTION NO: 259

What are two characteristics of OER? (Choose two.)

- A. It can take on HSRP, VRRP, and GLBP as clients.
- B. It provides automatic inbound route optimization.
- C. Path selection may be based on delay, loss, or jitter.
- D. The border router makes decisions about which outbound path to use.
- E. Automatic load distribution is provided for multiple connections.

Answer: C,E

Explanation:

QUESTION NO: 260

What are two characteristics of GSS? (Choose two.)

- A. It helps verify end-to-end path availability.
- B. It provides traffic rerouting in case of disaster.
- C. HSRP, GLBP, and VRRP can be clients of GSS.
- D. BGP must be the routing protocol between the distributed data centers.
- E. DNS responsiveness is improved by providing centralized domain management.

Answer: B,E

Explanation:

QUESTION NO: 261

What is the traditional mode for a firewall?

- A. routed mode
- B. context mode
- C. bridged mode
- D. transparent mode
- E. full security mode

Answer: A

Explanation:

QUESTION NO: 262

Which three of the following descriptions are true about the firewall modes? (Choose three.)

- A. Transparent mode is layer 2.
- B. Routed mode is layer 3.
- C. Routed mode has 1 IP address.
- D. Transparent mode has 1 IP address.

Answer: A,B,D

Explanation:

QUESTION NO: 263

Which two statements about an interface configured with the asr-group command are correct?

(Choose two.)

- A. The FWSM supports up to 16 asymmetric routing groups.
- B. If a matching packet is not found, the packet is dropped.
- C. Asymmetric routing of return traffic is enabled.
- D. If a matching packet is found, the Layer 3 header is rewritten.
- E. If a matching packet is found, the Layer 3 header is rewritten and the packet is forwarded to the

default gateway.

Answer: B,C

Explanation:

QUESTION NO: 264

Which two of these correctly describe asymmetric routing and firewalls? (Choose two.)

- A. only operational in routed mode
- B. only operational in transparent mode
- C. only eight interfaces can belong to an asymmetric routing group
- D. operational in both failover and non-failover configurations
- E. only operational when the firewall has been configured for failover

Answer: C,D

Explanation:

QUESTION NO: 265

In which two locations in an enterprise network can an IPS sensor be placed? (Choose two.)

- A. bridging VLANs on two switches
- B. bridging two VLANs on one switch
- C. between two Layer 2 devices with trunking
- D. between two Layer 2 devices without trunking
- E. between a Layer 2 device and a Layer 3 device with trunking

Answer: C,D

Explanation:

QUESTION NO: 266

Which three mechanisms are used to secure management traffic from outside IPS sensors? (Choose three.)

- A. secure tunnels

- B. a separate management VLAN
- C. secure VLANs to isolate sensors
- D. an out-of-band path around the firewall
- E. asymmetric traffic flows to isolate sensors
- F. private VLANs to put all sensors on isolated ports

Answer: A,B,F

Explanation:

QUESTION NO: 267

Which two statements about Cisco Security Management Suite are correct? (Choose two.)

- A. It should be implemented in a management VLAN.
- B. Its connection to managed devices should be over a data VLAN.
- C. It is made up of Cisco Security MARS and Clean Access software.
- D. It should be deployed as close to the edge of the network as possible.
- E. It delivers policy administration and enforcement for the Cisco Self-Defending Network.

Answer: A,E

Explanation:

QUESTION NO: 268

To ensure quality, what is the maximum end-to-end transit time in milliseconds on a voice network?

- A. 50
- B. 100
- C. 150
- D. 200
- E. 250

Answer: C

Explanation:

QUESTION NO: 269

Which three of these are elements of the Cisco Unified Wireless Network architecture? (Choose three)

- A. cell phones
- B. remote access
- C. mobility services
- D. network management
- E. network unification
- F. network decentralization

Answer: C,D,E

Explanation:

QUESTION NO: 270

For acceptable voice calls, the packet error rate should be no higher than what value?

- A. 0.1%
- B. 1%
- C. 2.5%
- D. 25%

Answer: B

Explanation:

QUESTION NO: 271

How many channels are defined in the IEEE 802.11b DSSS channel set?

- A. 3
- B. 4
- C. 11
- D. 13
- E. 14

Answer: E

Explanation:

QUESTION NO: 272

What amount of cell overlap ensures smooth roaming for wireless endpoints?

- A. 510%
- B. 1015%
- C. 1520%
- D. 2025%

Answer: C

Explanation:

QUESTION NO: 273

In a VoWLAN deployment, what is the recommended separation between cells with the same channel?

- A. 19 dBm
- B. 67 dBm
- C. 10 dBm
- D. 86 dBm
- E. 5 dbm to 10 dBm

Answer: A

Explanation:

QUESTION NO: 274

What is the recommended radius of a cell for a voice-ready wireless network?

- A. 6 dBm
- B. 19 dBm
- C. 5 dBm
- D. -67 dBm

Answer: D

Explanation:

At the edge of each voice cell, the received signal strength indication (RSSI) measurement should

be -67 dBm if you are using a Cisco Unified Wireless IP Phone 7921G. It is recommended that you have RSSI above 35 at the edge of the cell, which is equivalent to -67dBm for optimum performance on the phone.

Each cell in the network should overlap with the adjacent cells in order to facilitate uninterrupted handoff as a client moves between cells and to provide a minimum service even in case of access point failure. For a typical voice deployment, Cisco recommends a 15 to 20 percent overlap of a given access point's cell from each of the adjoining cells

QUESTION NO: 275

Client traffic is being bridged through LAN interfaces on two WLCs. Which roaming option will keep them on the same IP subnet?

- A.** Layer 1 intercontroller roaming
- B.** Layer 2 intercontroller roaming
- C.** Layer 3 intercontroller roaming
- D.** intracontroller roaming

Answer: B

Explanation:

QUESTION NO: 276

During consultation, you find that a customer has multiple asset closets and will be adding more in the future. Which NAS physical deployment model would you suggest to this customer?

- A.** edge
- B.** central
- C.** Layer 2
- D.** Layer 3

Answer: B

Explanation:

QUESTION NO: 277

The Cisco NAC Appliance is able to check which three items before allowing network access?

(Choose three.)

- A. client antivirus software state
- B. personal firewall settings
- C. wireless cell bandwidth availability
- D. IOS versions for routers and switches
- E. appropriate client patch management level
- F. appropriate QoS settings for client application

Answer: A,B,E

Explanation: